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UltraPACSrcDefTopic

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About the UltraPAC80 Source

About the UltraPAC80 Source

The Ultra Precision Analog Channel (UltraPAC80) Source provides AC waveform generation for applications such as wireless baseband, audio, HDTV/video, xDSL, DVD/ODD, and set top box/cable modem device testing. Typical tests that can be performed include: gain accuracy, frequency response, distortion, and signal-to-noise ratio, and linearity.

The following topics are included:

- Features
- Safety Information
- Key Terms
- UltraPAC80 Source Software Licensing Requirements

UltraPAC80 Source information includes the following sections:

- UltraPAC80 Source Theory of Operation
- UltraPAC80 Source Programming
- UltraSource Debug Display
- UltraPAC80 Source Examples

Related Topics

- UltraPAC80 Specifications
- UltraSource (VBT syntax)
- UltraPAC80 DIB Design Guidelines

Eight UltraPAC80 Source instruments reside on the UltraPAC80 instrument board with eight UltraPAC80 Capture instruments, which digitize AC waveforms. For more information on UltraPAC80 Capture, see About the UltraPAC80 Capture.

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About the UltraPAC80 Source : Features

Features

The UltraPAC80 Source provides:

- 8 fully independent sources
  - Independent user clocks and SMEM (32 Msamples per source)
  - Independent pattern control
  - Independent user-defined delay and amplitude control per channel (does not require reloading SMEM)
- High Frequency (HF) Mode DC to 80 MHz bandwidth
  - 400 MHz max sampling rate
  - Max AC swing 3Vpp single-ended, 6Vpp differential
- High Resolution (HR) Mode DC to 50 kHz bandwidth
  - Max AC swing 5Vpp single-ended, 10Vpp differential
- Single-ended and differential performance

- Better than 56 ps channel-to-channel skew without calibration DIB
- Sample rate conversion architecture
- Access to Parametric Measurement Unit (PPMU) per pin, 32 PPMUs per channel card
- Signals loading initiated from digital patterns
- Bidirectional operation from UltraSource out of UltraCapture for four channels per instrument board (two I/Q port pairs)
- Ability to assign each UltraPAC80 channel to a unique time domain (16 total, eight UltraSources and eight UltraCaptures)
- Ability to control each UltraPAC80 channel by unique time domains simultaneously using Sync-Link
  - Independent access to Sync-Link
  - Independent sample clock programming
  - Independent clock synchronization with other UltraPAC80 channels or with digital clocks
  - Independent control by digital pattern
  - Independent waveform memory and sequencing
- Pin-to-pin compatibility with BBAC and TurboAC
- BBAC-to-UltraPAC80 conversion tool to aid in test program porting
- Connections sets (CSets) that allow user defined connection setups to be applied using digital patterns

For more information, see UltraPAC80 Specifications.

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About the UltraPAC80 Source : Safety Information

Safety Information

When operating the UltraPAC80 Source instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety.

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About the UltraPAC80 Source : Key Terms

Key Terms

channel	A single Instrument to DIB connection. For example, a differential pair consists of two channels. Two single-ended sites consist of two channels.
common mode	The voltage added to both Pos and Neg signals by DC baseline hardware, referenced to DGS. In single-ended operation, the common mode DC baseline circuit can be used to provide a voltage offset to the waveform output.
DGS	Device Ground Sense. The DGS input is tied to the DUT's analog ground plane. This ensures that all signals sourced or captured from the UltraPAC80 are referenced to device ground rather than to instrument ground.
High Frequency mode	An instrument parameter that specifies use of the High Frequency (HF) signal path, DC - 80 MHz, when the 80MHz bandwidth is selected.
High Resolution mode	An instrument parameter that specifies use of the High Resolution (HR) signal path, DC - 50 kHz. This path is used when 1kHz and 50kHz bandwidths are selected.
IIM	Instrument Initiated Move
instrument	The set of UltraPAC80 functionality that can independently source or capture a waveform. An instrument includes functionality to generate and capture a waveform, as well as the functionality to deliver or receive the waveform from multiple outputs or inputs.

ISL	Instrument Sync-Link
ksps	Kilosamples per second
Msps	Megasamples per second
offset	The voltage added between Pos and Neg signals. Note: Offset is created using Sample Memory as opposed to hardware dc baseline circuitry (programmed with common mode).
PPMU	Pin Parametric Measurement Unit
SMEM	Source Memory

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About the UltraPAC80 Source : UltraPAC80 Source Software Licensing Requirements

UltraPAC80 Source Software Licensing Requirements

IG-XL enforces the following counted licenses for each UltraSource or UltraCapture channel (pos/neg channel pair):

UltraPAC80 License Enabling Points

License Name	Enabling Points
UltraSource	Waveform bandwidth in MHz: 80 (“full”), 40, 3
UltraCapture	Signal bandwidth in MHz: 75 (“full”), 40, 3

One license is required for each UltraSource or UltraCapture channel in the channel map, at a level that accommodates the highest signal frequency sourced or captured on the channel.

Online, IG-XL obtains these licenses from the counted license file. Offline, the simulated configuration file defines these counted licenses under the license enablers section.

In an offline simulated or current configuration file, these licenses appear as follows:

```
<LICENSE_ENABLERS>
#Off-line enabling levels
#FEATURE        ENABLED LEVEL    LICENSE COUNT    LEGAL ENABLING POINTS
UltraCapture    75               1                75,40,3
UltraSource     80               1                80,40,3
</LICENSE_ENABLERS>
```

IG-XL checks for UltraSource and UltraCapture licenses on validation of a test program to verify that the number of channels used in the channel map does not exceed the number licensed. If you do not have an adequate number of licenses for the number of channels, IG-XL returns an error. For example:

Unable to obtain enough UltraSource licenses to validate this job. At least 4 UltraSource licenses are required for channels in the Channel Map.

IG-XL also verifies licenses on execution of a test program to verify that the program does not source or capture signals whose frequency exceeds the licensed limitations. A run-time error occurs if the program sources or captures at frequencies higher than allowed by the licenses. For example:

Unable to obtain enough UltraCapture licenses to run this job. At least the following UltraCapture licenses are needed: 2 @ 3 MHz, 2 @ 40 MHz, 3 @ 75 MHz.

Each UltraCapture 75 MHz ("full") license allows one channel to operate at any legal capture sample rate. Each UltraCapture 3 MHz license allows one channel to operate up to a maximum sample rate of 8 Msps. This permits the channel to capture frequencies up to a maximum of 3 MHz, due to the requirement that the capture sample rate must be 8/3 times greater than the highest frequency of interest. Similarly, each UltraCapture 40 MHz license allows one channel to operate up to a maximum sample rate of (40 * 8/3) Msps, which permits the channel to capture frequencies up to a maximum of 40 MHz. See UltraPAC80 Specifications for the minimum sample rate specifications.

Each UltraSource 80 MHz ("full") license allows one channel to operate at any legal source sample rate. Each UltraSource 40 MHz license allows one channel to operate at any legal source sample rate up to a maximum of 160 Msps. This permits the channel to source frequencies up to a maximum of 40 MHz, due to the requirement that the source sample rate must be four times greater than the highest frequency of interest when the bandwidth setting is 80 MHz. Similarly, each UltraSource 3 MHz license allows one channel to operate at any legal source sample rate up to a maximum of 12 Msps. This permits the channel to source frequencies up to a maximum of 3 MHz. See UltraPAC80 Specifications for the minimum sample rate specifications.

For more information, see IG-XL Licensing in the IG-XL Administration and Licensing section.

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UltraPAC80 Source Examples

UltraPAC80 Source Examples

This section contains program examples for the UltraPAC80 Source instrument. These IG-XL workbook examples show all IG-XL data and VBT code necessary for a complete running program.

The following topics are available:

- UltraPAC80 Source/Capture Loopback
- UltraPAC80 Source Signals VBT Examples

For more information, see:

- UltraSource (VBT Syntax Reference)
- UltraPAC80 Source Programming

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UltraPAC80 Source Examples : UltraPAC80 Source/Capture Loopback

UltraPAC80 Source/Capture Loopback

This workbook example includes the following program components:

- DataTool Sheets
- VBT Code

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UltraPAC80 Source Examples : UltraPAC80 Source/Capture Loopback : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Test Instances Sheet
- Flow Table Sheet
- Time Sets (Basic)

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type	Comment
	Src1_P	Analog	
	Src1_N	Analog	
	Cap1_P	Analog	
	Cap1_N	Analog	
	DigControl	I/O	

Channel Map Sheet

Channel Map

DIB ID: **View Mode:** Signal

Device Under Test		Tester Channel		
Pin Name	**Package Pin**	**Type**	**Site 0**	**Comment**
Src1_P		UltraSource	22.SrcPos1	4
Src1_N		UltraSource	22.SrcNeg1	
Cap1_P		UltraCapture	22.CapPos1	
Cap1_N		UltraCapture	22.CapNeg1	
DigControl		I/O	1.ch0	

Test Instances Sheet

Test Instances

	Test Procedure			DC Specs		AC Specs		
Test Name	**Type**	**Name**	**Called As**	**Category**	**Selector**	**Category**	**Selector**	**Time Sets**
Source_Capture_LoopBack1	VBT	Source_Capture_LoopBack						TSB

Flow Table Sheet

Flow Table

		Gate			Command			
Label	**Enable**	**Job**	**Part**	**Env**	**Opcode**	**Parameter**	**TName**	**TNum**
					Test	Source_Capture_LoopBack1		

Time Sets (Basic)

Time Sets (Basic)

Timing Mode: Single **Master Timeset Name:**
Time Domain:

	Cycle	Pin/Group			Data		Drive				Compare			Edge Resolution
Time Set	**Period**	**Name**	**Clock Period**	**Setup**	**Src**	**Fmt**	**On**	**Data**	**Return**	**Off**	**Mode**	**Open**	**Close**	**Mode**
ts0	200.E-09	DigControl		i/o	PAT	NR		0			Off			Auto

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UltraPAC80 Source Examples : UltraPAC80 Source/Capture Loopback : VBT Code

VBT Code

```
Public Function Source_Capture_LoopBack()
    Dim SourceSine As New DSPWave
    Dim CaptureWave As New DSPWave
    Dim i As Long
    ' Connect UltraSource to UltraCapture Loopback
    TheHdw.UltraSource.Pins("Src1_P").Connect tlUltraSourceFrom_Pos Or _
        tlUltraSourceFrom_Neg, tlUltraSourceTo_Loopback
    TheHdw.UltraCapture.Pins("Cap1_P").Connect tlUltraCaptureConnectFrom_Pos _
        Or tlUltraCaptureConnectFrom_Neg, tlUltraCaptureConnectTo_Loopback
    ' Create Sinewave Samples to Source
    Call SourceSine.CreateConstant(0, 65536)
    For i = 0 To 65536 - 1
        SourceSine.Element(i) = Sin((3 * i * 2 * Pi / 65536)) / 2
    Next i
    ' Load Setting and Samples into the UltraPAC80 and Set up Source Signal
    TheHdw.UltraSource.Pins("Src1_P").Signals.Add "SourceSignal"
    TheHdw.UltraSource.Pins("Src1_P").Signals.DefaultSignal = "SourceSignal"
    TheExec.WaveDefinitions.CreateWaveDefinition "WaveDef", SourceSine, True
    With TheHdw.UltraSource.Pins("Src1_P").Signals("SourceSignal")
        .WaveDefinitionName = "WaveDef"
        .Amplitude = 1
        .Offset = 0
        .SampleSize = 65536
        .SampleRate = 655360
```

```
        .CycleCount = 1
        .LoadSettings
    End With
    ' Set up UltraCapture
    TheHdw.UltraCapture.Pins("Cap1_P").Signals.Add "CaptureSignal"
    TheHdw.UltraCapture.Pins("Cap1_P").Signals.DefaultSignal = "CaptureSignal"
    With TheHdw.UltraCapture.Pins("Cap1_P").Signals("CaptureSignal")
        .SampleSize = 65536
        .SampleRate = 655360
        .CycleCount = 1
        .LoadSettings
    End With
    ' Start Sourcing the signal and capturing with Pattern
    'TheHdw.Digital.ApplyLevelsTiming True, True, True
    TheHdw.Patterns("./src_cap_loop.pat").Load
    TheHdw.Digital.Patgen.TimeoutEnable = False
    TheHdw.Patterns("./src_cap_loop.pat").Start "", ""
    TheHdw.Digital.Patgen.HaltWait
    ' Wait for Capture to complete
    While TheHdw.UltraCapture.Pins("Cap1_P").IsCaptureInProgress
    Wend
    ' Bind capture data to a DSPwave object
    CaptureWave = TheHdw.UltraCapture.Pins("Cap1_P") _
        .Signals("CaptureSignal").DSPWave
    CaptureWave.Plot "UltraPAC80 LOOPBACK"
    ' Do an SNR Analysis on captured data and display it in immediate window
    Debug.Print CaptureWave.Spectrum.CalcSNR
    ' Disconnect the UltraSource from the UltraCapture
    TheHdw.UltraSource.Pins("Src1_P").Disconnect tlUltraSourceFrom_Pos Or _
        tlUltraSourceFrom_Neg, tlUltraSourceTo_Loopback
    TheHdw.UltraCapture.Pins("Cap1_P").Disconnect _
        tlUltraCaptureConnectFrom_Pos Or tlUltraCaptureConnectFrom_Neg, _
        tlUltraCaptureConnectTo_Loopback
End Function
```

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples

UltraPAC80 Source Signals VBT Examples

The following examples use UltraPAC80 Source VBT methods and properties:

- Apply Amplitude and Offset to an UltraPAC80 Source Ramp
- Apply Amplitude and Offset to an UltraPAC80 Source Sine Wave
- Create a Wave Definition for UltraPAC80 Source on the Fly
- Use MST to Set Up, Load, and Start an UltraPAC80 Source Signal
- Use UltraPAC80 Source Common Mode Hardware to Introduce an Offset
- Set Up and Start an UltraPAC80 Source Signal
- Use LoadSettings for an UltraPAC80 Source Signal and Read the Programmed Settings
- Use Pattern Control with UltraPAC80 Source Default Signal
- Report Information About a Collection of UltraPAC80 Source Signals

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples : Apply Amplitude and Offset to an UltraPAC80 Source Ramp

Apply Amplitude and Offset to an UltraPAC80 Source Ramp

This function applies amplitude and offset to a ramp wave definition.

```
Public Sub CreateUltraSourceRampSignal()
    ' This code creates a ramp for a single-ended input
    ' that goes from 2.5 V to 4.5 V.
    ' Replace "AIN" with your UltraPAC80 source (pos) pin name
    Dim pinName As String
    pinName = "AIN"
    ' Create a new Signal called "ramp1" on pin 'pinName'.
    TheHdw.UltraSource.pins(pinName).Signals.Add "ramp1"
    With TheHdw.UltraSource.pins(pinName).Signals("ramp1")
        ' Specify a single-ended device input:
        .ConnectionType = tlUltraSourceConnectionType_SingleEndedNeg
        ' Program WaveDefinitionName. Specify "aRamp" on the WaveDef sheet
        .WaveDefinitionName = "aRamp"
        ' Program Ampliutde to 2.0 - the difference between 4.5 and 2.5
        .Amplitude = 2#
        ' Program Offset to 2.5 - the level to start the ramp at.
        ' Note: example "Use CommonMode hardware to Introduce Offset"
        ' shows a way to use hardware to do this, which can improve
        ' signal quality by using more dynamic range of the converter.
        .Offset = 2.5
        ' Program SampleRate and SampleSize; CycleCount defaults to 1.
        .SampleRate = 20000#
```

```
        .SampleSize = 50
        ' View the samples to verify the Signal is set up.
        .samples.Plot
    End With
End Sub
```

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples : Apply Amplitude and Offset to an UltraPAC80 Source Sine Wave

Apply Amplitude and Offset to an UltraPAC80 Source Sine Wave

This function applies amplitude and offset to a sine wave definition for an UltraPAC80 Source.

```
Public Sub CreateUltraSourceUnipolarSineSignal()
    ' This creates a 1 kHz, 2 Vpk sine wave for a differential input
    ' that has a DC Offset of 2.5 V.
    ' Replace "AIN" with your UltraPAC80 source (pos) pin name
    Dim pinName As String
    pinName = "AIN"
    ' Create a new Signal called "sine1" on pin 'pinName'.
    TheHdw.UltraSource.pins(pinName).Signals.Add "sine1"
    With TheHdw.UltraSource.pins(pinName).Signals("sine1")
        ' ConnectionType defaults to Differential, so no need to set it.
        ' Program the WaveDefinitionName. Specify "aSine" on the WaveDef
        ' sheet
        .WaveDefinitionName = "aSine"
        ' Program Amplitude to 2.0. This is the difference between the
        ' DC level 2.5 and the peak (0.5 or 4.5).
        .Amplitude = 2#
        ' Program Common Mode Offset to 2.5. This is the DC level.
        .CommonMode = 2.5
        ' Program SampleRate and SampleSize; CycleCount defaults to 1.
        .SampleRate = 20000#
        .SampleSize = 20
```

```
        ' View the samples to verify the Signal is set up.
        .samples.Plot
    End With
End Sub
```

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples : Create a Wave Definition for UltraPAC80 Source on the Fly

Create a Wave Definition for UltraPAC80 Source on the Fly

This function uses CreateWaveDefinition to make a wave definition from an imported file on the fly.

```
Public Sub ImportFileForUltraSourceSignal()
    ' Use the FileImport function on the DSPWave to import a pre-existing file
    ' called "aCustom.txt". Create a wave definition from this file and
    ' assign it to Signal "customSignal" on pin "AIN".
    ' Replace "AIN" with your UltraPAC80 Source (pos) pin name
    Dim pinName As String
    pinName = "AIN"
    Dim mySamples As New DSPWave
    TheHdw.UltraSource.pins(pinName).Signals.Add "customSignal"
    With TheHdw.UltraSource.pins(pinName).Signals("customSignal")
        mySamples.FileImport "aCustom.txt", File_txt
        ' Use CreateWaveDefinition to define a wave definition
        ' from the imported samples on the fly.
        TheExec.WaveDefinitions.CreateWaveDefinition "myCustom", mySamples
        ' Set the signals wave definition to the wave definition just created
        .WaveDefinitionName = "myCustom"
        ' Set the SampleRate (SampleSize is inherent in the imported data)
        .SampleRate = 20000
        ' Amplitude and Offset are built into the wave definition, so use the
        ' defaults of 1.0 and 0.0 to apply no gain and offset.
        ' View the samples to verify the Signal is set up.
        .samples.Plot
```

```
    End With
End Sub
```

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples : Use MST to Set Up, Load, and Start an UltraPAC80 Source Signal

Use MST to Set Up, Load, and Start an UltraPAC80 Source Signal

This function uses a Mixed Signal Timing (MST) set to set up, load, and start a 2 Vpk differential source signal.

```
Public Sub UseMSTToSetupAndStartUltraSourceSignal()
    ' Use MST to set up, load, and start a 2 Vpk diff source signal.
    ' Replace "AIN" with your UltraPAC80 source (pos) pin name
    Dim pinName As String
    pinName = "AIN"
    TheHdw.UltraSource.pins(pinName).Signals.Add "sine1"
    With TheHdw.UltraSource.pins(pinName).Signals("sine1")
        ' Specify a Mixed Signal Timing Set for this Test Instance
        ' before calling ApplyMixedSignalTiming.
        .ApplyMixedSignalTiming
        ' Program Amplitude to 2 V pk.
        .Amplitude = 2#
        ' Use defaults for all other Signal properties
        ' Load Signal hardware values into hardware
        .LoadSettings
        ' Start sourcing the Signal
        .Start
    End With
End Sub
```

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples : Use UltraPAC80 Source Common Mode Hardware to Introduce an Offset

Use UltraPAC80 Source Common Mode Hardware to Introduce an Offset

This function uses CommonMode hardware to introduce an offset in a sine wave for a single-ended device input.

```
Public Sub UseCommonModeForSEUltraSourceSignal()
    ' Use CommonMode hardware to introduce offset in a sine wave
    ' for a single-ended device input.
    ' Replace "AIN" with your UltraPAC80 Source (pos) pin name
    Dim pinName As String
    pinName = "AIN"
    TheHdw.UltraSource.pins(pinName).Signals.Add "sine1"
    With TheHdw.UltraSource.pins(pinName).Signals("sine1")
        ' This creates a sine wave for a single-ended device input
        ' that goes from 0.5 V to 4.5 V.
        .ConnectionType = tlUltraSourceConnectionType_SingleEnded
        .WaveDefinitionName = "aSine"
        .Amplitude = 2#
        .CommonMode = 2.5
        ' Program SampleRate and SampleSize; CycleCount defaults to 1.
        .SampleRate = 20000
        .SampleSize = 20
        ' Load hardware Settings. This will load the CommonMode hardware.
        .LoadSettings
        ' Start sourcing
        .Start
```

```
    End With
    ' Read Back CommonMode hardware. Set it to 2.5.
    Debug.Print TheHdw.UltraSource.pins(pinName).CommonModeVoltage.Value
End Sub
```

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples : Set Up and Start an UltraPAC80 Source Signal

Set Up and Start an UltraPAC80 Source Signal

This function sets up and starts a signal.

```
Public Sub CreateAndStartUltraSourceSignalWithDirectProgramming()
   ' Use direct timing programming to set up, load, and start
   ' a 1 kHz, 2 Vpk diff source signal.
   ' Replace "AIN" with your UltraPAC80 Source (pos) pin name
   Dim pinName As String
   pinName = "AIN"
   TheHdw.UltraSource.pins(pinName).Signals.Add "sine1"
   With TheHdw.UltraSource.pins(pinName).Signals("sine1")
      .SampleRate = 20000
      .SampleSize = 20
      .CycleCount = 1   ' 1 is the default
      .WaveDefinitionName = "aSine"
      .Amplitude = 2
      ' Use defaults for all other Signal properties
      ' Load hardware settings for the Signal
      .LoadSettings
      ' Start the Signal
      .Start
   End With
End Sub
```

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples : Use LoadSettings for an UltraPAC80 Source Signal and Read the Programmed Settings

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Use LoadSettings for an UltraPAC80 Source Signal and Read the Programmed Settings
This function programs a new signal, uses Load Settings, and reads the programmed hardware settings.

```
Public Sub ProgramUltraSourceSignalAndReadbackHardware()
    ' Program a new signal, Load Settings, and read the programmed hardware
    ' settings from Pins language.
    ' Replace "AIN" with your UltraPAC80 Source (pos) pin name
    Dim pinName As String
    pinName = "AIN"
    TheHdw.UltraSource.pins(pinName).Signals.Add "sine1"
    With TheHdw.UltraSource.pins(pinName).Signals("sine1")
        .SampleRate = 20000
        .SampleSize = 20
        .CycleCount = 1
        .WaveDefinitionName = "aSine"
        .Amplitude = 2
        .VoltageRange.Mode = tlCSignalModeUseCalculatedValue
        .CommonMode.Value = 2.5
        ' Use defaults for everything else.
        ' Load hardware settings
        .LoadSettings
    End With
    ' Now that settings are loaded, read back hardware values
    ' from Pins language
```

```
    With TheHdw.UltraSource.pins(pinName)
        Debug.Print "Loaded SampleRate: " & CStr(.SampleRate)
        Debug.Print "Loaded VoltageRange: " & CStr(.VoltageRange)
        Debug.Print "Loaded CommonMode: " & CStr(.CommonModeVoltage)
    End With
End Sub
```

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples : Use Pattern Control with UltraPAC80 Source Default Signal

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Use Pattern Control with UltraPAC80 Source Default Signal

This function uses pattern control with the Default Signal.

```
Public Sub UseUltraSourceDefaultSignalAndPatternControl()
    ' Replace "AIN" with your UltraPAC80 Source (pos) pin name
    Dim pinName As String
    pinName = "AIN"
    ' Replace ".\Patterns\myPat.pat" with your pattern name
    Dim patternName As String
    patternName = ".\Patterns\myPat.pat"
    ' Add 5 Signals to the Signals collection on Pin 'pinName'.
    ' Give each signal an Amplitude from 0.5 to 2.5.
    Dim i As Long
    Dim sigCount As Long
    Dim sigList() As String
    For i = 1 To 5
        TheHdw.UltraSource.pins(pinName).Signals.Add "signal" & CStr(i)
        With TheHdw.UltraSource.pins(pinName).Signals("signal" & CStr(i))
            ' Specify a Mixed Signal Timing Set for this Test Instance
            ' before calling ApplyMixedSignalTiming.
            .ApplyMixedSignalTiming
            .Amplitude = 0.5 * i
        End With
    Next i
```

```
    ' Set each Signal to be the default signal and burst the pattern.
    For i = 1 To 5
        ' Load Settings
        TheHdw.UltraSource.pins(pinName).Signals("signal" & _
                                CStr(i)).LoadSettings
        ' Set the DefaultSignal
        TheHdw.UltraSource.pins(pinName).Signals.DefaultSignal = "signal" & _
                                CStr(i)
        ' This pattern should have an unnamed "Trig" microcode on pin
        ' 'pinName'. Run "" will start the pattern from the beginning and
        ' wait for completion.
        TheHdw.Digital.Patterns.Pat(patternName).Run ""
    Next i
End Sub
```

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UltraPAC80 Source Examples : UltraPAC80 Source Signals VBT Examples : Report Information About a Collection of UltraPAC80 Source Signals

Report Information About a Collection of UltraPAC80 Source Signals

This function reports information about a collection of UltraPAC80Source signals using Count and List.

```
Public Sub ReportInfoAboutUltraSourceSignalsColl()
    ' Replace "AIN" below with your UltraPAC80 source (pos) pin name
    Dim pinName As String
    pinName = "AIN"
    ' First, add 10 Signals to the Signals collection on Pin 'pinName'
    Dim i As Long
    Dim sigCount As Long
    For i = 1 To 10
        TheHdw.UltraSource.pins(pinName).Signals.Add "signal" & CStr(i)
    Next i
    sigCount = TheHdw.UltraSource.pins(pinName).Signals.Count
    ' Print each Signal name
    Dim sigList() As String
    sigList = TheHdw.UltraSource.pins(pinName).Signals.List
    Debug.Print "There are " & sigCount & " signals in the collection " & _
                "on pin '" & pinName & "'. They are:"
    For i = 0 To sigCount - 1
        Debug.Print vbCrLf & sigList(i)
    Next i
End Sub
```

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UltraSource Debug Display

UltraSource Debug Display

The UltraSource Debug Display is an interactive diagram of the UltraPAC80 Source hardware and signal path, which shows the instrument's settings and connections in the context of a site. You can change the values, options, and settings in the display and run your test program to see the effects of the changes.

All UltraPAC80 instruments are also displayed on the UltraPac Connections display.

The following topics are available:

- Starting the UltraSource Debug Display
- UltraSource Debug Display Interface
- Using the UltraSource Debug Display
- Using the UltraPac Connections Debug Display

This section describes features specific to the UltraSource debug display and the UltraPac Connections display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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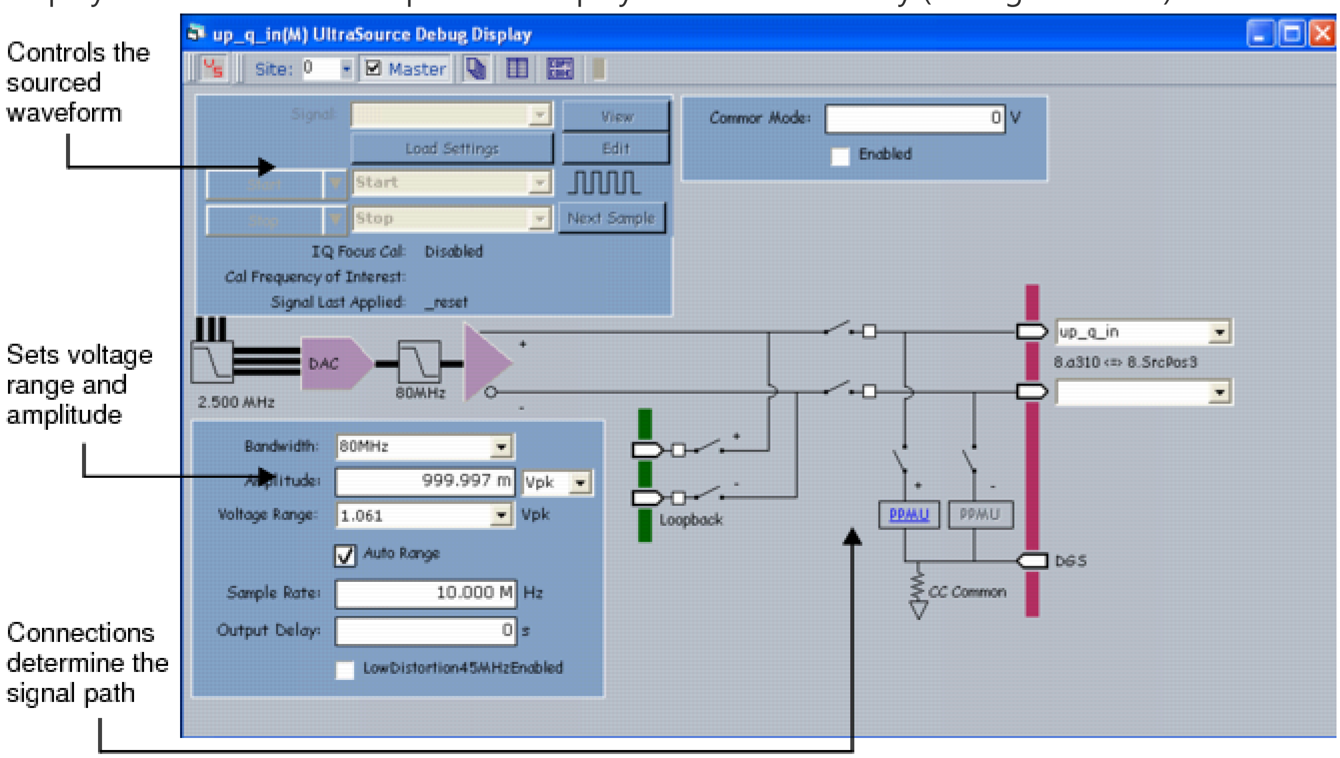
UltraSource Debug Display : Starting the UltraSource Debug Display

Starting the UltraSource Debug Display

To start the UltraSource debug display for an UltraPAC80 Source instrument, first validate an UltraPAC80 test program. Then, on the IG-XL Tools tab, in the Launch group, select Debug Displays>UltraSource. This opens the display for active sites only (see figure below).



UltraSource Debug Display

If you start a debug display before the job has been validated, an error appears in the display saying that the program has not been validated. If you open a display for an instrument that is in the configuration file but not in the channel map, the display shows a message that no instruments have been found.

Once the job is validated, the display shows the current hardware settings. When you run to a breakpoint in the flow or in the VBT code, or as you step through the flow or code, IG-XL updates the display to show any changes to the settings.

For information on other methods for starting debug displays, see Starting Debug Displays in the Test Debug Environment (TDE) section.

For information on the fields used in the UltraSource debug display, see UltraSource Debug Display Interface.

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UltraSource Debug Display : UltraSource Debug Display Interface

UltraSource Debug Display Interface

The following sections describe the interface for the UltraSource debug display:

- Controlling the Source Signal
- Setting the Voltage Range and Amplitude

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UltraSource Debug Display : UltraSource Debug Display Interface : Controlling the Source Signal

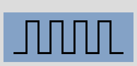
Controlling the Source Signal

The block in the upper-left section of the display includes input boxes and buttons that select which signal to source and where the source instrument will receive the start and stop commands from. You can also use the View button to preview the signal to be sourced in a scope display. Note that most fields and buttons only become active once a signal is specified in the Signal parameter.

The table lists the menus, buttons, and fields that control sourcing of signals in the UltraSource debug display.

UltraSource Waveform Debug Display Fields

Menu, Button, or Field	Action
Signal	The menu includes the names of the defined signals. If a default signal has been defined, the default signal's name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT.
View	Brings up the WaveScope tool and plots the sourced signal segment on the WaveScope display. This allows you to visually preview and analyze the signal that is to be sourced. When you are running the program on a test system, WaveScope displays the samples and scales them by the amplitude. When working offline, the WaveScope displays the ideal waveform.

Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Load Settings	Loads the signal’s timing settings as specified on the Mixed Signal Timing sheet, as well as all of the instrument’s settings (for example, voltage range, amplitude, and offset).
Start	This menu selects a type of start: • Start: Starts sourcing the signal associated with the specified signal according to the action command selected in the menu next to the Start button. The choices for start are: Start, StartE, Start1, or Start1E. • Multi-Pin Start: This selection allows you to start the signal for multiple pins.
Stop	This menu selects a type of stop: • Stop: Stops sourcing the signal associated with the specified signal according to the action command selected in the menu next to the Stop button. The choices are Immediately or At End. • Multi-Pin Stop: This selection allows you to stop the signal for multiple pins.
	The square wave is a sourcing indicator that moves when the source is sourcing on the tester. This feature only works on line.
Next Sample	Moves the instrument to move to the next sample when in single sample step mode.
IQ Focus Cal	This read-only field indicates whether IQ calibration is Enabled or Disabled.
Cal Frequency of Interest	This read-only field indicates the frequency of interest for calibration of a signal.
Signal Last Applied	This read-only field indicates the name of the last signal applied.

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UltraSource Debug Display : UltraSource Debug Display Interface : Setting the Voltage Range and Amplitude

Setting the Voltage Range and Amplitude

The block in the lower-left section of the display includes input boxes that allow you to select the voltage and amplitude settings of the output signal. You can also set the common mode voltage to center the AC signal around a common mode voltage, or you can set the Auto Range feature to automatically adjust the voltage range to source the signal within the largest dynamic range.

The table lists the menus, buttons, and fields that control the voltage and amplitude of signals in UltraSource debug display.

Voltage Range and Amplitude Display Fields

Menu, Button, or Field	Description
Bandwidth	The bandwidth of the source: 1kHz, 50kHz, or 80MHz.
Amplitude	Sets the amplitude of the signal as it will be seen by the DUT. • Minimum amplitude: 0.0 Vpk, differential • Maximum amplitude: 5.00 Vpk, differential
Voltage Range	Sets the voltage range of the signal. The selection of voltage ranges depends on the Bandwidth. • Minimum voltage range: 1.061 Vpk, differential

	• Maximum voltage range: 5.00 Vpk, differential
Auto Range	When selected, auto range mode is active. As the amplitude changes, the instrument automatically changes the voltage range so that it has the highest signal fidelity by using the entire dynamic range. Using Auto Range results in applying the smallest voltage range that can support the programmed amplitude. When auto range is deactivated, the voltage range is left at the last voltage range specified.
Common Mode	Sets the common mode voltage around which the AC signal is centered. • Minimum voltage: -5.0 V • Maximum voltage: +5.0 V
Enabled	Enables common mode voltage.
Sample Rate	Represents the number of samples to source per second. • Minimum sample rate: 5.00 ksps • Maximum sample rate: 400 Msps
Output Delay	Sets the delay in seconds for an UltraSource output path.
LowDistortion45MHzEnabled	Sets the UltraSource low distortion filter to improve performance of 45MHz sine waves. To use this option, Bandwidth must be set to 80MHz.

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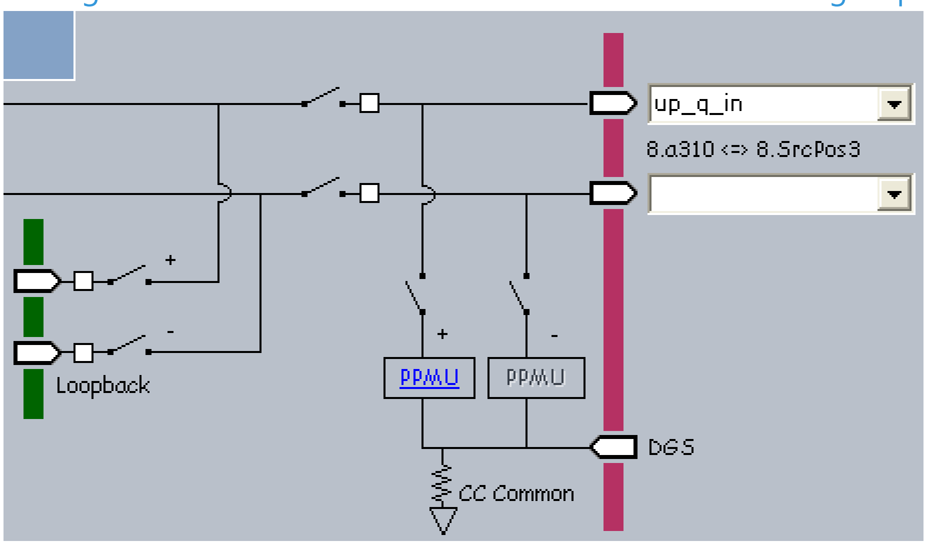
UltraSource Debug Display : Using the UltraSource Debug Display

Using the UltraSource Debug Display

The figure shows the connections on the UltraSource debug display.



UltraSource Debug Display Connections

The following topic describes the interface behavior and the actions that you perform to make channel and resource connections in the UltraSource debug display.

- Opening and Closing a Relay Path

The following topics describe the resources that the UltraPAC80 Source channels can be connected to:

- Connecting DUT Pins

- Connecting the Internal Analog Loopback Path

- Connecting a PPMU

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UltraSource Debug Display : Using the UltraSource Debug Display : Opening and Closing a Relay Path

Opening and Closing a Relay Path

The UltraSource debug display simulates channel connections to the DUT and other resources. Graphically, these connections are represented by open and closed relays along the connection path. Clicking a relay does not immediately open or close that relay.

To open or close a relay path:

1. Right-click the node along the signal path. A menu appears and lists the connect or disconnect option for that path.
2. Select the connection destination. The relay opens or closes to make or break the connection.

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UltraSource Debug Display : Using the UltraSource Debug Display : Connecting DUT Pins

Connecting DUT Pins

The UltraSource debug display allows you to connect an instrument channel to the DUT pins specified in the test program's channel map.

Starting the debug display automatically reads in the pins and lists them in the DUT port menus. The DIB channel string is also read into the debug display and is shown beside the appropriate instrument channel.

To connect an instrument channel or resource to a DUT pin:

1. Right-click the DUT node.
2. Select the channel from the menu.

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UltraSource Debug Display : Using the UltraSource Debug Display : Connecting the Internal Analog Loopback Path

Connecting the Internal Analog Loopback Path

The loopback connection is provided for debugging. It allows you to connect an UltraCapture directly to an UltraSource, allowing them to measure the sourced waveform.

The UltraPAC80 instrument board has eight instrument pairs (for example, UltraCapture and UltraSource in Channel 1). An UltraCapture Positive loops back to an UltraSource Positive Loopback is paired src/cap 1/2, 3/4, 5/6, and 7/8. Pos can only connect to Pos and Neg to Neg.

You can control the Pos and Neg channels independently. For example, you can loop back Positive or Negative or Both.

To connect an UltraCapture to an UltraSource for loopback purposes:

1.	Right-click the Loopback Positive node and select Connect to Source Pos.
2.	Right-click the Loopback Negative node and select Connect to Source Neg.
3.	Open the UltraCapture debug display.
4.	Right-click the Loopback Positive node and select Connect to Capture Pos.
5.	Right-click the Loopback Negative node and select Connect to Capture Neg.

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UltraSource Debug Display : Using the UltraSource Debug Display : Connecting a PPMU

Connecting a PPMU

You can connect each UltraSource channel to a Pin Parametric Measurement Unit (PPMU) circuit for performing continuity tests.

Although the UltraSource display shows the channel connections to the PPMU circuits, you make PPMU connections in the PPMU display, not in the UltraSource display.

To connect an UltraSource DUT pin (Pos or Neg) to a PPMU circuit:

1. Click the appropriate PPMU link PPMU on the UltraSource display based on the UltraSource channel that is to be connected. This PPMU debug display starts.
2. Select the pin to be connected to the PPMU from the Pin menu.
3. Click the relay to close the connection.

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UltraSource Debug Display : Using the UltraPac Connections Debug Display

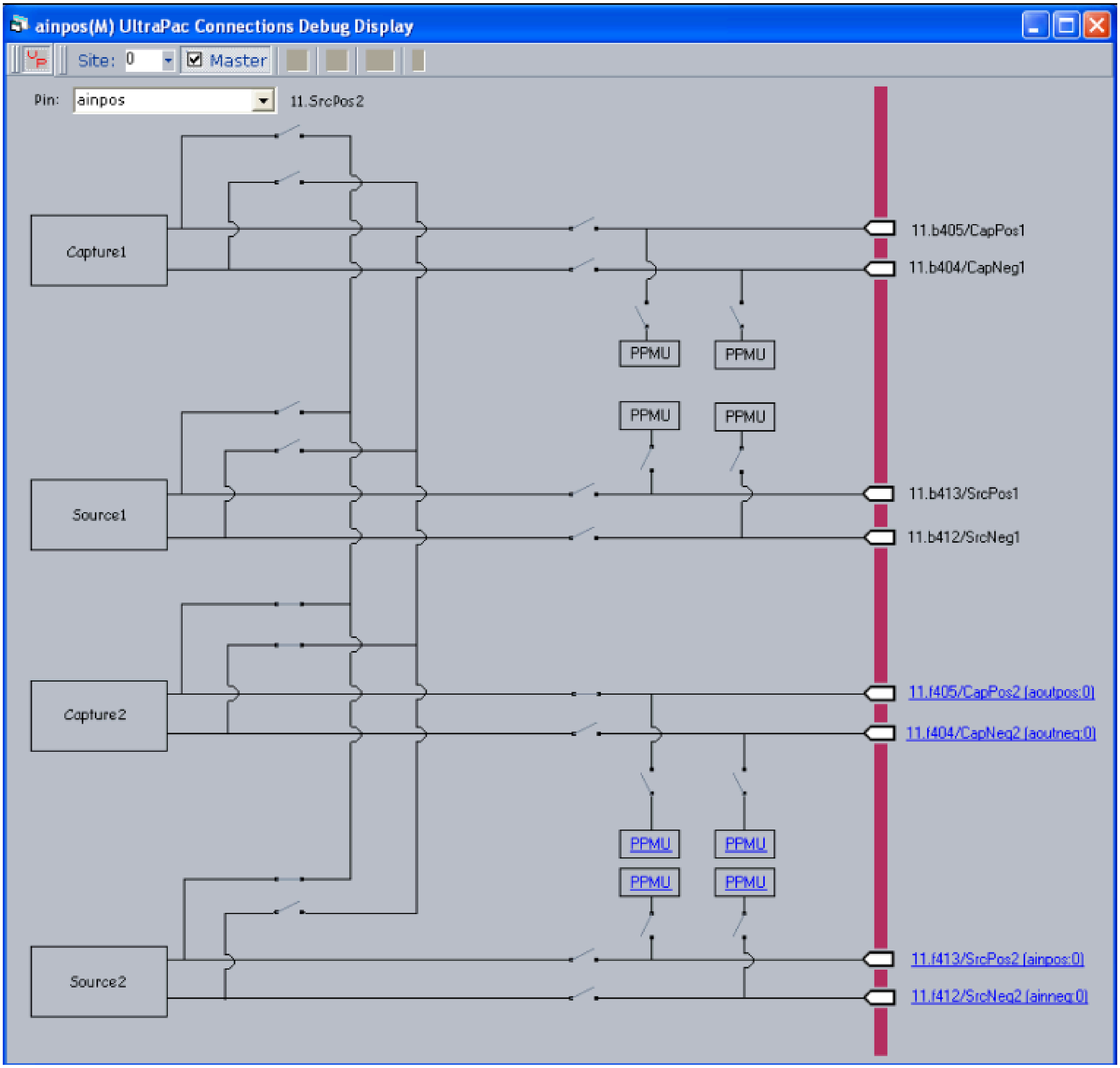
Using the UltraPac Connections Debug Display

The UltraPac Connections Debug Display is a read-only display showing boardwide connections.

The display shows an overview of UltraCapture, UltraSource, and PPMU connections.

Viewing the Display

To view this display, from the menu bar select Subsystems>UltraPac Connections in the Launch group on the IG-XL Tools tab. The figure shows an example of the display.



UltraPac Connections Debug Display

Using the Display

As a test is run, the display changes depending on the test program being executed. Channel information is highlighted on each instrument representation and relays open and close as the state of the instrument changes.

The display includes navigation links to channels that are mapped to pins. The link label is in following format: <Channel>(<Pin>:<site>). Click an instrument link to open the debug display for that instrument.

If an UltraSource is connected in bidirectional mode, then the port at source and at the capture will have links to both the UltraSource and UltraCapture debug displays.

The display also has static links to PPMU display.

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UltraPAC80 Source Programming

UltraPAC80 Source Programming

This section describes programming of an UltraSource instrument to source data to a DUT.

The following topics are available:

- Simulated Configuration Files
- Programming Concepts
- Programming Overview
- UltraSource DataTool Programming
- UltraSource Pattern Control
- Using UltraPAC80 in Concurrent Testing
- Achieving IQ Alignment for UltraSource
- Programming the Relay Counter
- Using the BBAC to UltraPAC Converter Tool

For more information, see:

- UltraSource (VBT Syntax Reference)
- UltraPAC80 Source Examples

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UltraPAC80 Source Programming : Simulated Configuration Files

Simulated Configuration Files

You can use an UltraSource instrument offline by forcing IG-XL to map the UltraPAC80. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps an UltraPAC80 board in slot 15:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]   Type            idprom (type serial rev company)

15.0        UltraPAC        605-743-00 0000000 0000-A 5445
            UltraPAC        939-420-01 0000000 0000-A 5445
            UltraPAC-0      939-392-00 0000000 0000-A 5445
            UltraPAC-1      939-392-00 0000000 0000-A 5445
            PowerBoard      604-207-00 0000000 0000-A 5445

24.0        SupportBoard    974-220-12 0000000 0000-A 0000
```

SupportBoard 956-354-00 0000000 0000-A 0000

62.0 DistributionBoard 956-355-00 0000000 0000-A 0000
DistributionBoard 956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>

To use an UltraPAC80 Source offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the UltraSource online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see Tester Configuration.

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UltraPAC80 Source Programming : Programming Concepts

Programming Concepts

The following topics are available:

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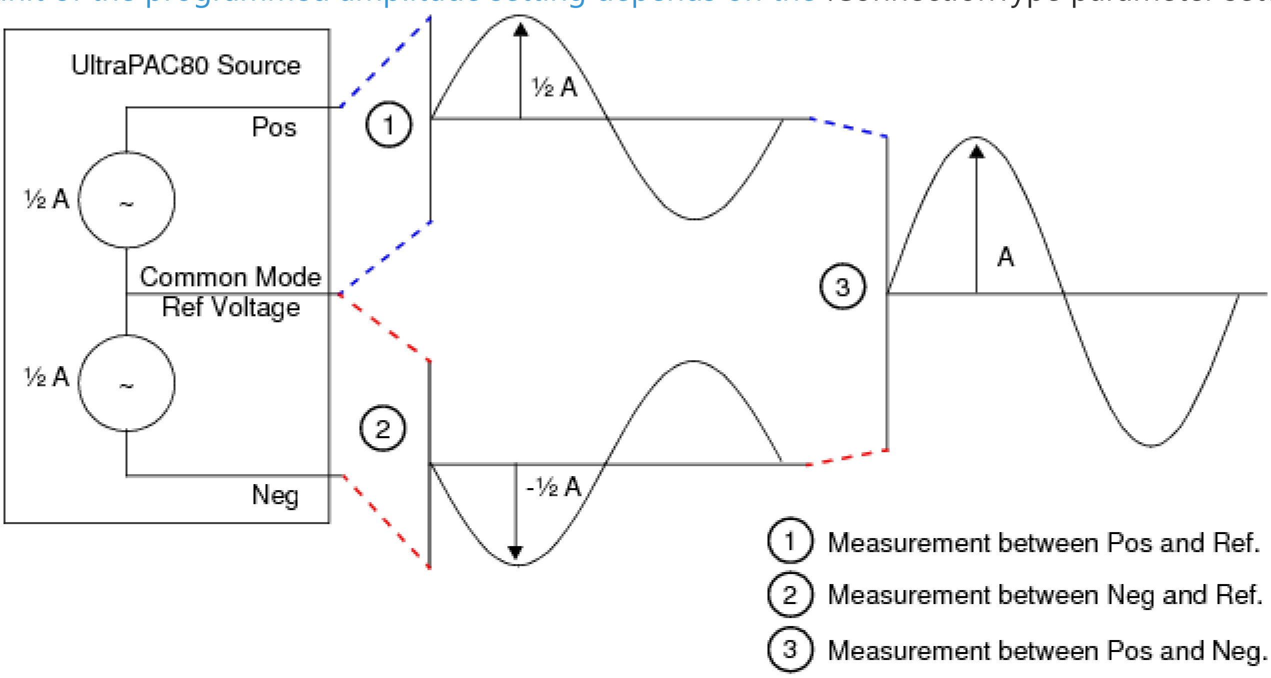
UltraPAC80 Source Programming : Programming Concepts : Differential and Single-Ended Measurements

Differential and Single-Ended Measurements

The figure shows an UltraSource instrument where "A" is the programmed amplitude. The scale or unit of the programmed amplitude setting depends on the .ConnectionType parameter setting.



UltraSource Range Measurements

If .ConnectionType is set to tlUltraSourceConnectionType_Differential, then the amplitude represents the peak voltage of a bipolar waveform measured differentially across the positive (Pos) and negative (Neg) output pins. If the peak differential voltage is A, then the peak voltage (Vpk) is A when measured single-ended to ground on only one of the differential output pins, either pos or neg.

When using an UltraSource in a single-ended output configuration, set .ConnectionType to tlUltraSourceConnectionType_SingleEnded. This scales the programmed amplitude such that if the

amplitude is programmed to A, then the peak amplitude observed on either the pos or neg output pin relative to ground will be A, not A as is with the differential setting.
Likewise, peak-to-peak voltage (Vpp) is defined as the maximum voltage minus the minimum voltage of the output waveform. The peak-to-peak voltage observed is always twice the peak programmed amplitude.
The UltraSource range specifies the maximum or minimum voltage that is generated between the positive and negative channels, which is equivalent to a measurement in Vpk, differential. Ranges are always expressed in units peak differential, regardless of the connection type setting.
For example, if the amplitude is set to 1.2V and the connection type is differential, then the range must be set to a value greater than or equal to 1.2V. If the amplitude is set to 1.2V and the connection type is single-ended, then the range must be set to a value >= 2 x 1.2V = 2.4V. Setting the range to a value smaller than the amplitude is invalid.
UltraSource ranges are:

- High Resolution mode:313mV, 442mV, 625mV, 884mV, 1.250V, 1.768V, 2.500V, 3.536V, and 5.000V
- High Frequency mode: 530mV, 750mV, 1.061V, 1.500V, 2.121V, and 3.000V

The UltraSource includes an autorange feature, which automatically adjusts the voltage range depending on amplitude and connection type. To use autoranging, set the Signals.VoltageRange.Mode property to tlCSignalModeUseCalculatedValue.
You can operate an UltraSource in a single step mode. In this mode source memory samples are incremented one at a time, asynchronously using microcode instead of using the sample clock at a fixed sample rate. To operate an UltraSource in this mode, set the sample rate to zero. This instructs the instrument to look for the NextSample pattern microcode or the Signals.NextSample VBT command instead of using the sample clock to advance through each sample of the corresponding signal.
For more information, see:

- TheHdw.UltraSource.Pins.ConnectionType
- TheHdw.UltraSource.Pins.Signals.Item.VoltageRange.ConnectionType
- TheHdw.UltraSource.Pins.Signals.NextSample

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UltraPAC80 Source Programming : Programming Concepts : Internal Analog Loopback

Internal Analog Loopback

The UltraPAC80 includes four loopback paths, allowing you to source and capture signals between instruments on the board. The following groups of instruments share a single loopback path:

- UltraSource 1 and 2 and UltraCapture 1 and 2
- UltraSource 3 and 4 and UltraCapture 3 and 4
- UltraSource 5 and 6 and UltraCapture 5 and 6
- UltraSource 7 and 8 and UltraCapture 7 and 8

You can use the UltraPAC Connections debug display to show loopback connections. This loopback feature can be used for verifying instrument performance and setup conditions without requiring a DIB or external instrumentation.

The following code demonstrates connecting an UltraSource to UltraCapture in loopback:

```
Public Function CloseUltraPACLoopbackRelay(SrcPin As String, _
                        CapPin As String) As Long
  TheHdw.UltraCapture.Pins(CapPin).Connect tlUltraCaptureFrom_Inst, _
                        tlUltraCaptureTo_LoopBack
  TheHdw.UltraSource.Pins(SrcPin).Connect tlUltraSourceFrom_Inst, _
                        tlUltraSourceTo_LoopBack
End Function
```

The following code demonstrates disconnecting an UltraSource from UltraCapture in loopback:

```
Public Function OpenUltraPACLoopbackRelay(SrcPin As String, _
                          CapPin As String) As Long
  TheHdw.UltraCapture(CapPin).Disconnect tlUltraCaptureFrom_Inst, _
                          tlUltraCaptureTo_LoopBack
  TheHdw.UltraSource(SrcPin).Disconnect tlUltraSourceFrom_Inst, _
                          tlUltraSourceTo_LoopBack
End Function
```

☑ **Note:**

Make sure that the UltraSource and UltraCapture are disconnected from other signal paths (for example, from the DIB/DUT) to avoid external signal interference while using internal loopback.

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UltraPAC80 Source Programming : Programming Concepts : Setting Bandwidth

Setting Bandwidth

You can set UltraSource instruments to the following bandwidths:

tlUltraSourceBandwidth__1kHz	1 kHz bandwidth through the high resolution signal path.
tlUltraSourceBandwidth__50kHz	50 kHz bandwidth through the high resolution signal path.
tlUltraSourceBandwidth__80MHz	80 MHz bandwidth through the high frequency signal path. Default.

The bandwidths have different ranges for SampleRate, CommonMode, and VoltageRange. Compared to the high frequency signal path, the high resolution signal path has:

- A lower maximum sample rate
- Higher common mode voltage
- Higher voltage ranges

Because of the exclusive nature of SampleRate, CommonMode, and VoltageRange with respect to bandwidth, a change in bandwidth may change the state of other parameters. When switching between the bandwidths, IG-XL does not return an error if the currently programmed sample rate, common mode, or voltage range is beyond the range of the newly programmed bandwidth. Instead, IG-XL reprograms the properties to the smallest or largest values valid in that bandwidth. The Bandwidth parameter is unique in this respect. Other VBT parameters return errors if a programmed state is invalid because of another parameter’s state.

To eliminate invalid transitional instrument states and improve throughput, use the signals method of programming rather than specifying individual settings using .Pins programming.

The .LowDistortion45MHzEnabled property is set to false when changing bandwidth modes.

For more information, see:

- Bandwidth (VBT Syntax Reference)
- UltraPAC80 Specifications

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UltraPAC80 Source Programming : Programming Concepts : Synchronous Triggering of Multiple Instruments

Synchronous Triggering of Multiple Instruments

To synchronize triggering of multiple instruments you can use the synchronizeChannels option of the .Start method. If you specify synchronizeChannels=True, IG-XL starts all selected instruments synchronously. This is analogous to programming the Resync and Start microcodes using pattern control, except that this method does not require pattern control to achieve synchronization between UltraPAC source channels. However, subsequent changes to the sample rate (using .SampleRate or .LoadSettings) will disrupt this synchronization.

Note the following:

- To use this method you must specify multiple pin names in a single .Signals.Start command. Therefore, you must use the same signal name for every pin that is to be started simultaneously. Individual signal properties may be different on each pin, but the signal name must be the same.

- If synchronizeChannels=True, IG-XL automatically issues a settlewait before starting the source. This ensures that I/Q calibration is adjusted properly and settled before the sources are started.

For more information, see Start in the VBT Syntax Reference.

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UltraPAC80 Source Programming : Programming Concepts : UltraSource Reset Values

UltraSource Reset Values

The table lists the local reset values for the UltraSource. Pre-job reset, post-job reset, power-up reset, jobload reset, and user reset all result in the same state, unless otherwise noted.

Reset Values for UltraSource Settings

Setting	Reset Value
Amplitude	1.0 Vpk
Bandwidth	80MHz
Common Mode Voltage	0.0 V
Connection Type	Differential
45 MHz Low Distortion Mode	False
Microcode Control	tlOff
Output Delay	0s
Sample Rate	10MHz
Voltage Range	1.0605 Vpk (autorange for 1.0V)

Reset Values for UltraSource Signal Settings

Setting	Reset Value
Calibration Frequency	1kHz

IQEnabled	False
Cycle Count	1
Offset	0
Phase	0

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UltraPAC80 Source Programming : Programming Concepts : Setting Distortion (Linearity) Calibration Range

Setting Distortion (Linearity) Calibration Range

UltraPAC80 instruments (UltraSource and UltraCapture) use a linearity calibration lookup table to improve harmonic distortion over a wide range of fundamental frequencies and amplitudes. A number of tables are available for different selections of bandwidth, voltage range, connection types, and frequency of interest to be sourced.

Since IG-XL does not know the frequency of interest, you must supply this information so that the correct calibration table is applied. To specify the frequency of interest, set the property Signals.Item.Calibration.FrequencyOfInterest to the Fi that it to be sourced given that signal setup. For multitone or complex waveforms, set .FrequencyOfInterest to the highest frequency (highest tone) that is to be sourced.

For more information, see FrequencyOfInterest in the VBT Syntax Reference.

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UltraPAC80 Source Programming : Programming Concepts : Connection Sets

Connection Sets

A connection set (CSet) is a named setup for UltraPAC80 instruments. CSets are defined in VBT and then executed from a pattern or from VBT to connect and disconnect UltraSources and UltraCaptures.

CSets provide the following advantages:

•	Fast connection times
•	Transparent test program modularity
•	No need to disconnect at the end of every test
•	Initiate every instrument setup entirely from a pattern burst

CSets are a more efficient means to set an instrument's connections. They provide higher throughput and require less relay switching, while still maintaining test independence.

A program that uses CSets for connection programming need not disconnect the instrument from the DUT at the end of each test to maintain test independence. Each test can use a connection set to ensure the connections it needs are made without regard to the previous state of the instrument.

Creating a New Connection Set

To add a CSet, use the ConnectionSets.Add syntax. For example, to add a new CSet called MyConnect, you would use:

TheHdw.UltraSource.Pins().ConnectionSets.Add("MyConnect")

Defining a Connection Set

You define a CSet by adding connections to an existing Connection Set. When you add connections, you can define the origin of each connection and its destination. Origins can include:

- Positive and negative pins
- PPMU positive and negative pins

The destination of all connections in a CSet is the DUT.

For example, for the UltraSource assigned to SecondPin, add a new connection to the CSet MyConnect from the positive pin to the DUT:

```
TheHdw.UltraSource.Pins("SecondPin").ConnectionSets.Item("MyConnect"). _
    AddConnection(tlUltraSourceCSetFrom_Pos, tlUltraSourceCSetTo_DUT)
```

You can call this method multiple times on one CSet to build up a list of connections.

CSets are an alternative to the Pins().Connect syntax. When applying a CSet, current connections are not considered. The UltraSource instrument handles any disconnects and other instrument operations to ensure that the hardware ends up in the specified state.

IG-XL provides the following predefined CSets for UltraSource:

ToDUT	On the selected instruments, disconnects any existing connections and connects the selected instruments to the DUT.
Disconnect	On the selected instruments, disconnects any existing connections. Used by the Disconnect pattern microcode

You cannot modify these CSets.

Applying a Connection Set

You apply CSets using either VBT syntax or using microcode from a pattern.

From a pattern, use the Connect microcode. For example, to apply the predefined CSet ToDUT, the command would be:

```
Connect ToDUT
```

On the selected instruments, this command disconnects any existing connections and connects the instruments to the DUT.

You can also apply CSets using VBT language. For example:

```
' Apply a CSET to make the connection
TheHdw.UltraSource.Pins("SrcPosI, SrcPosQ") _
    .ConnectionSets("CSET_DIFF_TO_DUT").Apply
```

IG-XL provides additional syntax that allows you to control and monitor CSets.

Connections Set VBT Example

The following is a CSets programming example:

```
' Create Differential CSET
With TheHdw.UltraSource.Pins("SrcPosI, SrcPosQ").ConnectionSets _
    .Add("CSET_DIFF_TO_DUT")
    .AddConnection tlUltraSourceCSetFrom_Pos, tlUltraSourceCSetTo_DUT
    .AddConnection tlUltraSourceCSetFrom_Neg, tlUltraSourceCSetTo_DUT
End With
' Create Single Ended POS CSET + PPMU on neg
With TheHdw.UltraSource.Pins("SrcPosI, SrcPosQ").ConnectionSets.Add("CSET_SE_AND_PPMU") _
    .AddConnection tlUltraSourceCSetFrom_Pos, tlUltraSourceCSetTo_DUT
    .AddConnection tlUltraSourceCSetFrom_PPMUNeg, tlUltraSourceCSetTo_DUT
End With
' Apply a CSET to make the connection
TheHdw.UltraSource.Pins("SrcPosI, SrcPosQ"). _
    ConnectionSets("CSET_DIFF_TO_DUT").Apply
```

For more information, see Pins.ConnectionSets in the VBT Syntax Reference.

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UltraPAC80 Source Programming : Programming Overview

Programming Overview

To develop a test program that uses an UltraSource instrument, follow this basic procedure:

1. Create a pin map with pins of the DUT (Programming the Pin Map)
2. Create a channel map that allocates the tester resources to the DUT pins (Programming the Channel Map).
3. Create a mixed signal timing set (Creating a Wave Definition and Mixed Signal Timing Set).
4. Create a test flow, the order in which the test will execute, in the Flow Table sheet.
   - Create a test function using VBT code in the Visual Basic editor.
   - Instantiate the VBT test and set the test values in the Instance Editor.
5. Validate and run the test program.

For more information, see UltraSource in the VBT Syntax Reference.

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UltraPAC80 Source Programming : UltraSource DataTool Programming

UltraSource DataTool Programming

This section provides instructions on setting up parameters in worksheets specific to the UltraSource. The following topics are included:

- Programming the Pin Map
- Programming the Channel Map
- Creating a Wave Definition and Mixed Signal Timing Set

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UltraPAC80 Source Programming : UltraSource DataTool Programming : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name.

For the UltraSource, the connections that you enter in the pin map are the analog channel connections to the device under test. Select the Analog pin type for each device pin name connected to an UltraSource instrument. Analog is the only valid pin type for UltraSource pins.

For more information, see Pin Map Sheet in the DataTool section.

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UltraPAC80 Source Programming : UltraSource DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For an UltraSource instrument, select the UltraSource channel type from the Type column menu for each pin or pin group.

You must enter the tester channel information value in the Site column for each site to connect a DUT pin to the instrument. The table lists the available UltraSource channels. Note that in the table, x indicates the DIB slot number of the UltraSource.

UltraSource Channel Resources

UltraSource Pogo Pin	DIB Channel	Signal Name
UltraSource Pos 1	x.b413	x.srcpos1
UltraSource Neg 1	x.b412	x.srcneg1
UltraSource Pos 2	x.f413	x.srcpos2
UltraSource Neg 2	x.f412	x.srcneg2
UltraSource Pos 3	x.b413	x.srcpos3
UltraSource Neg 3	x.b412	x.srcneg3
UltraSource Pos 4	x.f413	x.srcpos4
UltraSource Neg 4	x.f412	x.srcneg4
UltraSource Pos 5	x.b413	x.srcpos5
UltraSource Neg 5	x.b412	x.srcneg5

UltraSource Pos 6	x.f413	x.srcpos6
UltraSource Neg 6	x.f412	x.srcneg6
UltraSource Pos 7	x.b413	x.srcpos7
UltraSource Neg 7	x.b412	x.srcneg7
UltraSource Pos 8	x.f413	x.srcpos87
UltraSource Neg 8	x.f412	x.srcneg8

DGS pins need not be called out in a pin map or a channel map. DGS pins are automatically handled by the UltraPAC instrument when other connections are made.

You can also use the UltraPAC80 to make bidirectional connections. That is you can use an UltraSource channel through an UltraCapture channel's interposer. For bidirectional connections, select UltraSource from the Type menu for each Source channel and select the corresponding Capture channel's DIB channel or Signal name from the drop-down menu. For a list of UltraCapture channel resources, see Programming the Channel Map in the UltraCapture Programming section.

For more information, see:

- Channel Map Sheet
- Connectivity Options
- Bidirectional Connections

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UltraPAC80 Source Programming : UltraSource DataTool Programming : Creating a Wave Definition and Mixed Signal Timing Set

Creating a Wave Definition and Mixed Signal Timing Set

An UltraSource instrument generates and delivers analog signals of a specific shape, frequency, and amplitude to the DUT. These signals start as wave definitions, which define the basic shape of the signal. Then a timing solution is added to the waveform definition to create a timing set for the test. An UltraSource instrument uses the timing set to create a reusable signal to which various amplitudes can be applied.

The following topics provide instructions for creating a wave definition and a timing set:

- Setting Up the Wave Definitions Sheet
- Creating a Timing Set Using the Mixed Signal Workshop
- Calculating Source Timing
- Source Timing Example: 100 Hz in High Resolution Mode
- Source Timing Example: 1.5 MHz in High Frequency Mode
- Source Timing Example: 30 MHz in High Frequency Mode
- Loading Signals

Setting Up the Wave Definitions Sheet

A Wave Definitions sheet contains the wave shape information for waveform definitions that were created using the WaveDesigner. A Wave Definitions sheet stores the wave definition information and the test program accesses the information when the test is run.

For more information, see Waveform Definitions and About WaveDesigner in the WaveDesigner section.

Creating a Timing Set Using the Mixed Signal Workshop

The Mixed Signal Workshop (MSW) tool helps you to build reusable mixed signal timing solutions that meet coherence requirements.

The WaveDesigner tool creates wave definitions, which are independent of timing and instrument specific information. The MSW uses these wave definitions and then applies the instrument specific timing information necessary to generate samples.

The output of the MSW is a mixed signal timing set which contains the signal's structural characteristics and the details of the clocking solution that you created. Timing sets are stored on a Mixed Signal Timing sheet in the IG-XL workbook and become a reusable timing solution for future tests.

To add an UltraSource to the MSW display:

1.	Select MSW>Add Source.

2.	Select UltraSource from the Select Resource dialog box. Leave the Resource ID box empty, unless this timing set is using additional UltraPAC80 Source instruments. If that is the case, assign the current source instrument a unique ID name to distinguish it from the others. Note that the Resource ID must begin with a letter and can only contain letters, numbers, or the underscore character.

3.	Click OK in the Select Resource dialog box. The UltraPAC80 Source resource block is added to the MSW display.

For more information, see Mixed Signal Workshop.

Calculating Source Timing

Source timing solutions must adhere to the UltraSource timing rules:

•	High Resolution mode. The minimum Fs allowed is the larger of 5kHz or 10 times the highest frequency of interest:

$F_s >= \max[5kHz \,||\, 10 \times F_r]$

- High Frequency mode. Due to digital filtering in the resampler DSP, an UltraSource requires a source sample rate that is the larger of 5kHz or four times the highest frequency of interest:

F_s >= max[5kHz || 4 x F_r]

An UltraSource has minimum sample size requirements required for proper hardware operation. The minimum sample size for any waveform is 20. Additionally the length (time) of an individual waveform segment must be greater than or equal to 1s:

(3–1) $N \geq \max[20 \text{ or } \mathrm{ceiling}(1\mu s \times F_s)]$

This means that you cannot create a sinusoid with cylecount (M) = 1 if the Fi is greater than 1MHz.

Source Timing Example: 100 Hz in High Resolution Mode

The following example shows the source timing solution calculation for a 100Hz sine wave in High Resolution mode.

Fs >= max [5 kHz || 10 x Fi]

Fs >= max [5 kHz || 10 x 100 Hz]

Fs >= max [5 kHz || 1 kHz]

Fs >= 5 kHz

N >= max [20 || ceiling(1 s x Fs)]

N >= max [20 || ceiling(1 s x 5ksps)]

N >= max [20 || 0.005], the ceiling of 0.005 = 1

N >= 20

With Fs = 5 kHz and M = 1:

N = M x Fs/Fi

N = 1 x 5 kHz/100 Hz

N = 50 meets minimum requirement

Therefore, a valid timing solution for a 100 Hz sine wave in High Resolution mode is:

Fs = 5kHz

N = 50

M = 1

Source Timing Example: 1.5 MHz in High Frequency Mode

The following example shows the source timing solution calculation for a 1.5MHz sine wave in High Frequency mode.

Fs >= max [5 kHz || 4 x Fi]

Fs >= max [5 kHz || 4 x 1.5 MHz]

Fs >= max [5 kHz || 6 MHz]

Fs >= 6 MHz

N >= max [20 || ceiling(1 s x Fs)]

N >= max [20 || ceiling(1 s x 6Msps)]

N >= max [20 || 6]

N >= 20

With Fs = 6 MHz and M = 1:

N = M x Fs/Fi

N = 1 x 6 MHz/1.5 MHz

N = 4 does not meet the minimum requirement

So, change M to 5. With Fs = 6 MHz and M = 5:

N = M x Fs/Fi

N = 5 x 6 MHz/1.5 MHz

N = 20 meets the minimum requirement

Therefore, a valid timing solution for a 1.5MHz sine wave in High Frequency mode is:

Fs = 6MHz

N = 50

M = 5

Source Timing Example: 30 MHz in High Frequency Mode

The following example shows the source timing solution calculation for a 30MHz sine wave in High Frequency mode.

Fs >= max [5 kHz || 4 x Fi]

Fs >= max [5 kHz || 4 x 30 MHz]

Fs >= max [5 kHz || 120 MHz]

Fs >= 120 MHz

N >= max [20 || ceiling(1 s x Fs)]

N >= max [20 || ceiling(1 s x 120Msps)]

N >= max [20 || 120]

N >= 120

With Fs = 120 MHz and M = 1:

N = M x Fs/Fi

N = 1 x 120 MHz/30 MHz

N = 4 does not meet the minimum requirement

So, change M to 30. With Fs = 120 MHz and M = 30:

N = M x Fs/Fi

N = 30 x 120 MHz/30 MHz

N = 120 meets the minimum requirement

Therefore, a valid timing solution for a 30MHz sine wave in High Frequency mode is:

Fs = 120MHz

N = 120

M = 30

Loading Signals

You can load signals for an UltraSource using:

- VBT .LoadSettings syntax

- LoadSettings Microcode in a pattern

The process is as follows:

1. Define a signal and its parameters for an UltraSource in a VBT statement. This creates a signal data structure on the host PC.

For example, the following statement defines signals for UltraSource and UltraCapture:

```
' Set up the UltraPAC Source Signal
With TheHdw.UltraSource(SrcPin_I & ", " & SrcPin_Q) .Signals _
        .Add("BasicSrcSig")
    .ApplyMixedSignalTiming
    .Bandwidth = tlUltraSourceBandwidth__80MHz
    .Amplitude = SrcAmp
    .ConnectionType = tlUltraSourceConnectionType_Differential
    .CommonMode.Value = 1.5
    .CommonMode.Enabled = True
    .LoadSettings
End With
' Set up the UltraPAC Capture Signal
With TheHdw.UltraCapture(CapPin_I & ", " & CapPin_Q).Signals _
        .Add("BasicCapSig")
    .ApplyMixedSignalTiming
    .Bandwidth = tlUltraCaptureBandwidth__75MHz
    .Termination = tlUltraCaptureTermination_Term50
    .VoltageRange = SrcAmp
    .Offset.Enable = tlUltraCaptureOffsetEnable_PosAndNegEnable
    .Offset.Value = 1.5
    .LoadSettings
End With
```

2. Issue a command to load the signal by doing either of the following:

– Use the VBT command .LoadSettings

– Use the LoadSettings microcode in a pattern

For example, the following statement loads the signal MySignal:
.Signals("MySig").LoadSettings
IG-XL evaluates the signal to see if it is already loaded.

3. If the signal is not loaded, it is loaded into the signals memory of the UltraPAC80 channel card.

If the signal is already loaded, this step is skipped.

4. The hardware registers of the channel card are updated using a download from signals memory.

The evaluation of whether a signal is already downloaded, the signal download, and the update of hardware registers all add to test time. Obviously, updating hardware for a signal that is already in signals memory is faster than loading a new signal. However, .LoadSettings and the LoadSettings microcode still require that IG-XL evaluate a signal to see if it is new and unlike other signals in signals memory.
The VBT command .ReloadSettings allows you to eliminate the time to evaluate a signal. This command does not write any settings to signal memory, and it does not check the signal software data structure to see if it has changed. The command simply executes a download from signals memory to the channel card hardware registers.
For more information, see:

- LoadSettings
- ReloadSettings

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UltraPAC80 Source Programming : UltraSource Pattern Control

UltraSource Pattern Control

You can control an UltraSource instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling the instrument during pattern execution.

The following topics are available:

- UltraSource Pattern Microcodes
- Understanding UltraPAC Pattern Interlocking
- UltraSource ASCII Pattern Syntax

☑ **Note:**

To use pattern microcodes to control an UltraSource on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraPAC80 Source Programming : UltraSource Pattern Control : UltraSource Pattern Microcodes

UltraSource Pattern Microcodes

A microcode controls a single function, such a starting or stopping sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as digital channels) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used to issue commands to an UltraSource instrument from a pattern.

UltraSource Pattern Microcodes

Microcode	Functional Description
Resync	Forces phase alignment of an UltraSource instrument clock with the HSD T0 clock.
Start <SignalName>	The instrument immediately (on the next available logical analog clock) starts sourcing the named signal in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used. See TheHdw.UltraSource.Pins.Signals.DefaultSignal.
StartE <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the specified signal in SMEM and repeats it

	continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used. See TheHdw.UltraSource.Pins.Signals.DefaultSignal.
Start1 <SignalName>	The instrument immediately starts sourcing (on the next available logical analog clock) the specified signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used. See TheHdw.UltraSource.Pins.Signals.DefaultSignal.
Start1E <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the specified signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced.If the signal name is not specified, the default signal is used. See TheHdw.UltraSource.Pins.Signals.DefaultSignal.
Stop	The instrument stops sourcing the currently sourced signal immediately. Stop does not affect a signal that is running once.
StopE	The instrument stops sourcing the currently sourced signal at the end of the signal.
LoadSettings <SignalName>	Instructs an UltraSource to load the settings for the specified source signal. SignalName is optional. The default signal is used if no SignalName is included.
Connect <Connection Set_Name>	Instructs an UltraSource to switch its connections to the specified connection set. Predefined ConnectionSets are: • ToDUT: On the selected instruments, disconnects any existing connections and connects the instruments to the DUT.

	• Disconnect: On the selected instruments, disconnects any existing connections. See Connection Sets.
Disconnect	Instructs an UltraSource to reset all connections to their disconnected state.
NextSample	Move to the next sample in single sample step mode (when SampleRate has been programmed to 0). See Differential and Single-Ended Measurements.

☑ **Note:**

UltraPAC80 microcodes have specific rules regarding the necessary spacing in vectors between microcodes on a specific channel. See UltraPAC80 Specifications.

UltraSource does not use PSets. However, it provides the equivalent capability using Signals programming in VBT and the LoadSettings microcode in a pattern.

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UltraPAC80 Source Programming : UltraSource Pattern Control : Understanding UltraPAC Pattern Interlocking

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Understanding UltraPAC Pattern Interlocking

By default, hardware for UltraPAC instruments (UltraCaptures and UltraSources) is set to listen to programming commands that are initiated from VBT and that originate from the host computer. When a digital pattern is burst that contains UltraPAC microcodes, such as LoadSettings and Connect, the hardware is activated to also listen for instructions that originate from the digital pattern generator which are sent through the Instrument Sync Link (ISL) bus.

While an UltraPAC instrument is set to listen for ISL instructions, IG-XL locks out all commands originating from VBT, and a run-time error is generated if you attempt to program the UltraPAC instrument from VBT while the pattern is running. This pattern interlocking is done to avoid collisions in the hardware in the case where instructions from different sources occur too close together for the hardware to react properly to both instructions.

The lockout starts when a digital pattern containing LoadSettings and Connect or Disconnect microcodes is started. The lockout is released at end of a Digital.Patgen.Haltwait VBT statement indicating that the pattern burst is complete or when an explicit Patgen.Halt method is executed.

At times, however, a test program may be written such that VBT commands are to be issued to an UltraPAC instrument while a pattern is running. For example, you might need to issue commands while a pattern is looping on a wait for CPU flag. In such cases, you can override the pattern lockout to allow VBT programming to occur. The syntax is:

TheHdw.UltraSource.Pins("MyPin").PatternMicroCodeControl = tlOff

Once the needed VBT commands are issued, you should restore .PatternMicroCodeControl to tlOn before continuing pattern execution.

Note that changing the .PatternMicroCodeControl property only toggles the VBT interlock mechanism. It does not disable ISL commands. Collisions are still possible when

.PatternMicroCodeControl is set to tlOff and the digital pattern is still running and executing microcodes. Take care when toggling .PatternMicroCodeControl such that this does not occur.

For more information, see PatternMicroCodeControl in the VBT Syntax Reference.

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UltraPAC80 Source Programming : UltraSource Pattern Control : UltraSource ASCII Pattern Syntax

UltraSource ASCII Pattern Syntax

UltraSource pattern microcodes are instructions you can place on an individual vector of a pattern. For that vector, a microcode instructs the UltraSource to perform a particular action, such as starting a new signal.

You can only use UltraSource microcodes with pins that the instruments statement has specified as UltraSource. For example:

```
instruments = {
      DUT_AIN:UltraSource;
}
```

For more information, see Instruments Statement in the Pattern Language section.

If a vector contains any UltraSource related microcode other than nop, that vector is placed in SRM.

The format for specifying an UltraSource microcode on a particular vector is:

(pin-list : UltraSource = microcode)

The following code shows ASCII pattern microcode syntax for UltraSource and UltraCapture instruments. The device analog input (DUT_AIN) and analog output (DUT_AOUT) are connected to an UltraSource and an UltraCapture respectively.

```
instruments = {
      DUT_AOUT:UltraCapture;
      DUT_AIN:UltraSource;
}
vector ( $tset, Digital1, Digital2)
{
 > tset1  0       1       ;
 (DUT_AOUT = Resync)
 (DUT_AIN = Resync)
```

```
> -    1    1    ;
(DUT_AOUT = Trig )
(DUT_AIN = Start )
> -    0    1    ; /* Trigger/Start the default signal */
(DUT_AOUT = Trig Audio_1kHz)
(DUT_AIN = Start Audio_1kHz)
> -    1    1    ; /* Trigger/Start a named signal */
(DUT_AOUT = Enable_Alarm)
(DUT_AIN = Start1 )
> -    0    1    ;
(DUT_AOUT = Disable_Alarm)
(DUT_AIN = StartE )
> -    1    1    ;
(DUT_AOUT = Enable_Inst_Cond)
(DUT_AIN = Start1E )
> -    0    -    ;
(DUT_AOUT = Disable_Inst_Cond)
(DUT_AIN = Stop)
> -    1    -    ;
(DUT_AIN = StopE)
> -    0    -    ;
(DUT_AOUT = LoadSettings Cap_Signal)
(DUT_AIN = LoadSettings Src_Signal)
> -    0    -    ;
(DUT_AOUT = Connect DIFF_to_DUT)
(DUT_AIN = Connect DIFF_to_DUT)
> -    1    -    ;
> -    -    -    ;
}
```

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UltraPAC80 Source Programming : Using UltraPAC80 in Concurrent Testing

Using UltraPAC80 in Concurrent Testing

To support concurrent testing, you can assign each UltraPAC80 instrument channel (UltraCapture or UltraSource) to a unique time domain. Concurrent features for UltraPAC80 include:

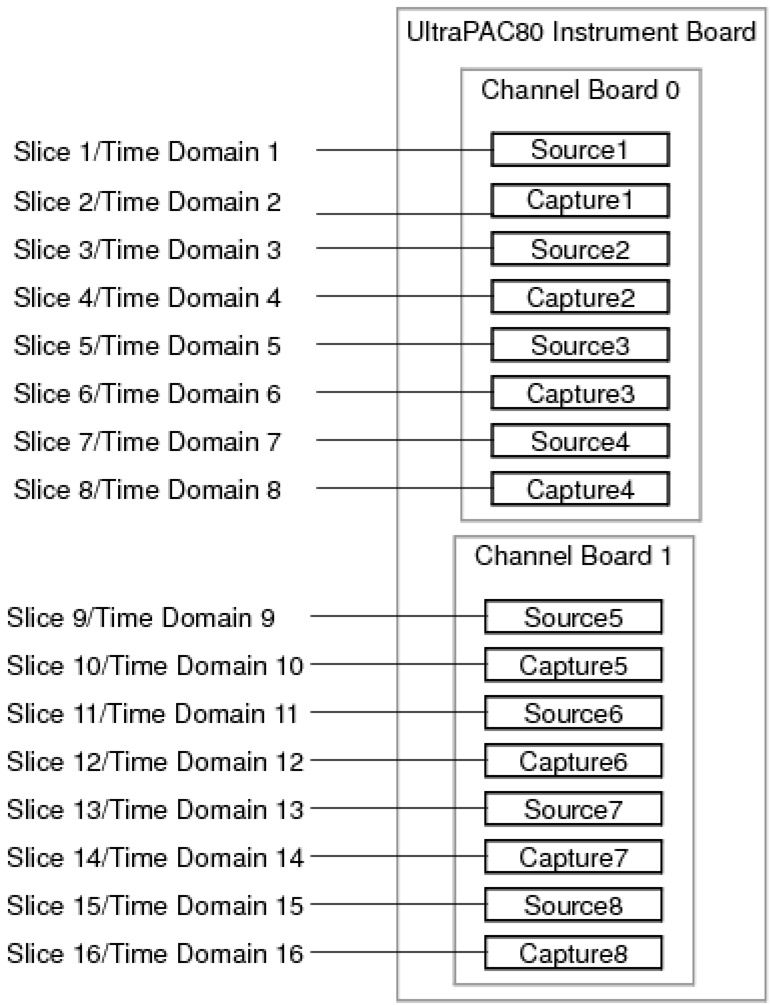
- Up to 16 instruments per channel card can be configured in up to 16 time domains:
  - Eight UltraSource instruments per channel card
  - Eight UltraCapture instruments per channel card
- Each UltraPAC80 instrument can be controlled by unique time domains simultaneously using Sync-Link. Features include:
  - Independent access to Sync-Link
  - Independent sample clock programming
  - Independent clock synchronization with other UltraPAC80 instruments or with digital clocks
  - Independent control by digital pattern
  - Independent capture memory and triggers

- Multiple instrument channels can be grouped into a common flow domain

- UltraPAC80 fully supports flow domains and asynchronous pattern starts

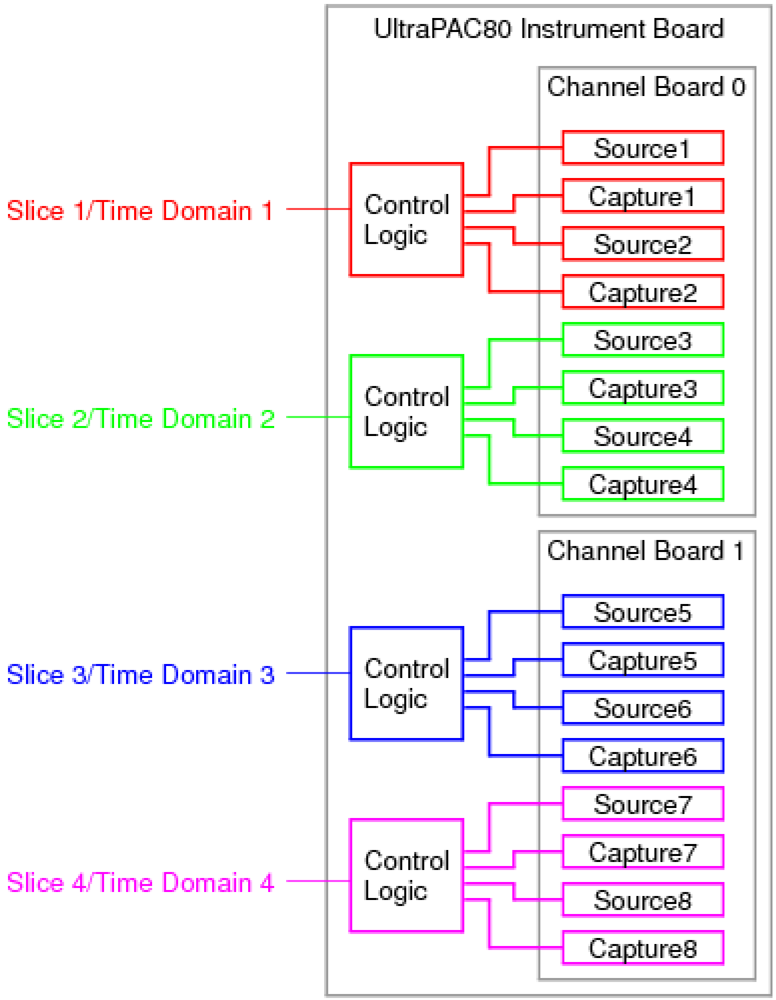
Note that when a shared UltraPAC80 resource is used within multiple concurrently executed flows, only one flow can communicate with the shared resource at a given time. Only the pattern associated with the time domain for which the ApplyLevelsTiming was last applied will be “active” and capable of communicating with the UltraPAC channel through the Instrument Sync Link (ISL), and only if the UltraPAC channel is also assigned to that time domain. When ApplyLevelsTiming is executed and a MOSC change occurs, IG-XL resolves that time domain and sets up appropriate ISL linkages between digital channels/PatGen, MOSC, and analog instrumentation that are all in the same time domain. If a pattern bust associated with a different time domain, which is not currently the “active” time domain, attempts to communicate with an UltraPAC instrument, then a BaseIslacHardware timeout error is returned. If this happens, adjust the sequencing of patterns and ApplyLevelsTiming calls such that the most recently executed ApplyLevelsTiming sets the correct time domain for the subsequent pattern burst.

The following figure shows a configuration in each UltraPAC80 instrument assigned to its own time domain, providing the maximum of 16 time domains.

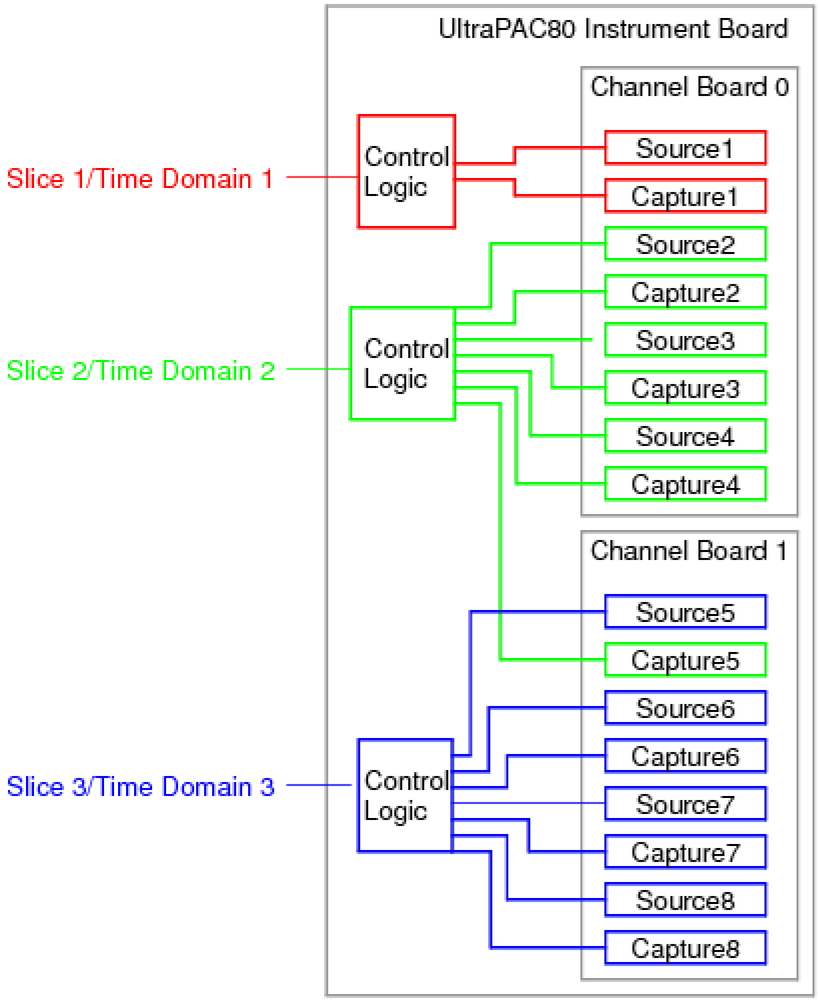


UltraPAC80 Concurrent Testing Configuration Example, 16 Time Domains

However, you can also group several UltraPAC80 instruments into a common time domain. The following figures show examples of UltraPAC80 instruments configured for four and for three time domains.



UltraPAC80 Concurrent Testing Configuration Example, Four Time Domains



UltraPAC80 Concurrent Testing Configuration Example, Three Time Domains

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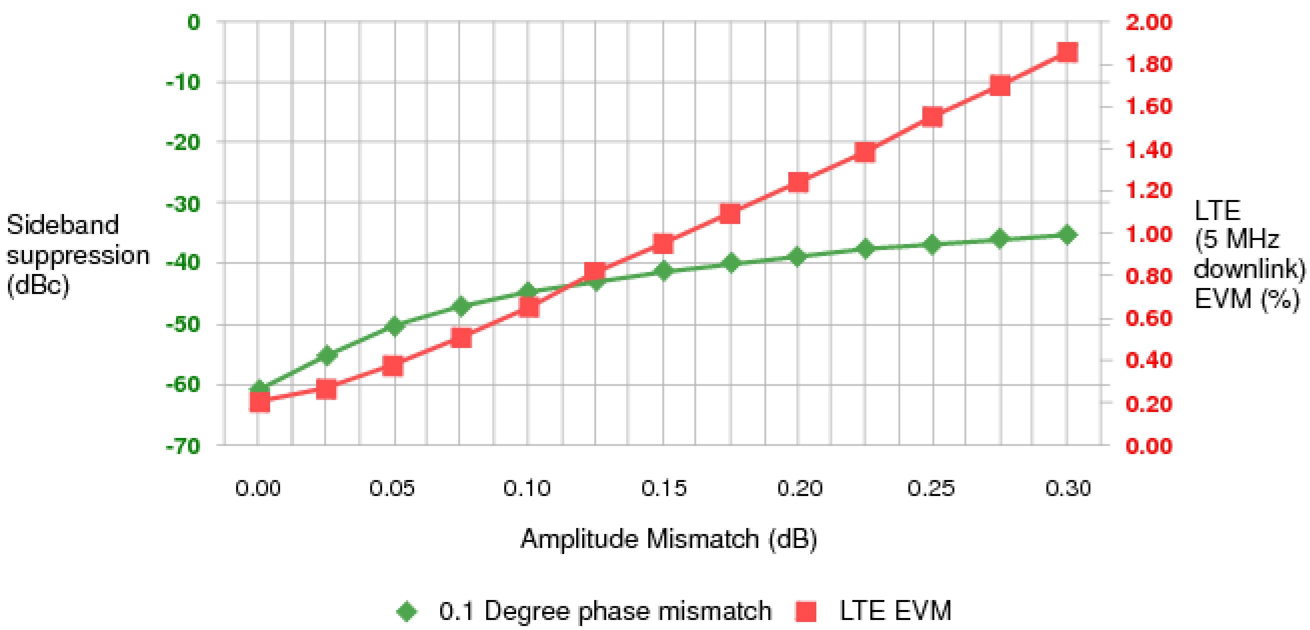
UltraPAC80 Source Programming : Achieving IQ Alignment for UltraSource

Achieving IQ Alignment for UltraSource

Poor sideband suppression degrades modulation metrics in wireless testing, and sideband suppression is completely determined by the alignment of in-phase and quadrature signal components (IQ alignment).

As an example, the following graph shows the relation of sideband suppression and amplitude mismatch to error vector magnitude (EVM) for LTE. As the graph shows, increasing amplitude mismatch rapidly decreases sideband suppression.



Sideband Suppression and LTE EVM for 0.1 Degree Phase Mismatch

The following topics are available:

- IQ Calibration Method
- Sourcing IQ Signals

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UltraPAC80 Source Programming : Achieving IQ Alignment for UltraSource : IQ Calibration Method

IQ Calibration Method

An UltraPAC80 instrument achieves IQ alignment using two types of calibration:

- Factory calibration stored on the instrument board
- Run-time IQ calibration performed when a signal is loaded

Factory Calibration for IQ Alignment

Factory calibration for IQ alignment is done by Teradyne and is fixed when an UltraPAC80 instrument is produced. The calibration values are stored on the EEPROMs for each instrument board. This calibration provides rough phase alignment (to within a few hundred picoseconds)

If IQ calibration is enabled when you load a signal, IG-XL determines whether that signal has been calibrated. If not, IQ calibration is performed, and factory calibration values (along with real-time values) are applied at that time.

Run-Time IQ Calibration

IG-XL includes a focused calibration routine that you can use on a per signal basis. This IQ calibration provides a final adjustment to the phase specification to account for components that vary over time and temperature, along with amplitude alignment.

You can run IQ calibration and apply it to any signal. If enabled, IQ focused calibration is performed at Signals.Item().LoadSettings or at pattern start, where the pattern contains a LoadSettings microcode for the signal.

By default, IG-XL checks to see if IQ focused calibration was previously executed for the specified signal on the instrument pair since job load. If calibration was executed, IG-XL applies the amplitude and phase correction factors to the source instrument pair for this signal. Otherwise, IG-

XL starts IQ calibration to phase-match the pair of instruments and amplitude-match the instruments to the signal's voltage range and calibration frequency of interest.

To use run-time IQ calibration, the channels that make up an I/Q pair must be adjacent. You can use the following channel pairs: 1/2, 3/4, 5/6, or 7/8. I/Q channel pairs that do not adhere to this rule return a run-time error when you try load a signal with IQ calibration enabled. See the example in Sourcing IQ Signals.

Note the following:

- IQ calibration only applies to the .Bandwidth and .VoltageRange parameters specified during a Signal's .LoadSettings. Modifying these parameters after .LoadSettings is applied results in degraded alignment.
- IQ calibration uses the frequency setting to obtain optimal alignment for that frequency. Instrument phase is adjusted for optimal alignment.
- No IQ matching is done during DSP preprocessing.
- Running the IQ calibration uses the associated UltraCapture instruments. Each UltraCapture instrument is restored when the run-time calibration is complete. However, the UltraCapture must not be running when .LoadSettings is called on the UltraSource.

For more information, see UltraSource in the VBT Syntax Reference.

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UltraPAC80 Source Programming : Achieving IQ Alignment for UltraSource : Sourcing IQ Signals

Sourcing IQ Signals

The following is an example programming procedure for sourcing an IQ signal. This procedure uses two sources, one for the I signal and one for the Q signal, that are phase-matched and amplitude-matched

The procedure is as follows:

1. From a loaded test program, define two pins for the I and Q signals. For example, the channel map could be:

Example IQ Channel Map

Pin Name	Type	Site 0	Comments
IPin	UltraSource	<slot>.srcpos1	1st source instrument
QPin	UltraSource	<slot>.srcpos2	2nd source instrument

2. Define a signal, enabling IQ calibration to phase-match and amplitude-match the two source instruments to source the signals:

```
With TheHdw.UltraSource.Pins("IPin, QPin").Signals.Add("IQSource")
  ' <Settings of instrument>
  .VoltageRange = <Signal voltage range> ' Used for amplitude matching
  .SampleRate = <Signal sample rate> ' Program together to align sources
  .Calibration.Enabled = True
```

```
  .Calibration.FrequencyOfInterest = <Fi of signal to be sourced in Hz>
End With
```

3.	Load the instrument settings of the Signal:

```
TheHdw.UltraSource.Pins("IPin, QPin").Signals("IQSource").LoadSettings
```

Once the settings are loaded, IG-XL determines if IQ focused calibration was previously executed for this signal on the instrument pair since this job load. If so, IG-XL applies the amplitude and phase correction factors to the source instrument pair for this signal. Otherwise, IG-XL starts IQ calibration to phase-match the pair of instruments and amplitude-match the instruments to the specified calibration frequency of interest and signal's voltage range.

After IQ calibration, all user settings and connections are restored to their previous states.

4.	Connect the instruments to the DUT:

```
TheHdw.UltraSource.Pins("IPin, QPin").Connect
```

5.	Source the I and Q signals:

```
TheHdw.UltraSource.Pins("IPin, QPin").Signals("IQSource"). _
   Start(tlStartImmediately, True)
```

IG-XL sources the two source instruments synchronously.

For more information, see UltraSource in the VBT Syntax Reference.

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UltraPAC80 Source Programming : Programming the Relay Counter

Programming the Relay Counter

The mechanical relays used in the UltraSource have a finite life span and can perform a limited number of switches before they begin to fail. To maximize the life span of the instrument, you may want to minimize the number of times a relay is switched during a test program run. IG-XL includes a RelayCounter feature for the UltraSource that allows you to collect relay switching data.

All relay switches are counted whether initiated from the tester computer using VBT or from the pattern with a microcode.

Relay counting is unaffected by resets (those before or after a test program or resets you initiate), but counts do not persist beyond the current program. Loading a new test program or restarting DataTool resets relay switching counts. Counts are enabled by default.

For example, you have a test program loaded that includes two UltraSource pins (Src1 and Src2) with two sites each. You want to determine how many times each UltraSource relay switched during a program run. You can do the following:

1.	Open an immediate window and executes code to begin the relay counting:

```
TheHdw.UltraSource.Pins("Src1, Src2").RelayCounter.Disabled = False
TheHdw.UltraSource.Pins("Src1, Src2").RelayCounter.Start
```

IG-XL initializes or resets its user relay counting variables.

Note that this is a convenient time to start relay counting, but generally no sites are active or selected at this time. This method (and all RelayCounter features) ignores site status and selection.

2.	Run the test program or loop it several times.

IG-XL counts all switches of mechanical relays from all source actions (calibration, reset, pattern, and tester computer).

3.	When the test program completes its runs, execute code in the immediate window a to generate a relay counting report and save it to a text file:

TheHdw.UltraSource.Pins("Src1, Src2").RelayCounter.DumpCurrentCount "UltraSource_Relay.txt"

Each count includes the number of times a relay has switched since the Start call in step 1, including auto and run-time calibrations.

IG-XL then generates a formatted text file that includes all relays associated with the UltraSource (in rows) and data for each pin/site specified in the pins selection (in columns) regardless of site activity/selection. The report is formatted in a fixed-width font and tab delimited to allow you to paste it into an Excel worksheet.

Here is an example of a relay report showing four UltraSource instruments:

```
******************* Dump Relay Counter for UltraSource **********************

Pin             src1      src1      src2      src2
Site            0         1         0         1
Slot.Chan#      2.SrcPos1 2.SrcPos2 2.SrcPos3 2.SrcPos4

CONNECT_HR_POS        0   0   0   0
CONNECT_HR_NEG        0   0   0   0
CONNECT_HF_POS        0   0   0   0
CONNECT_HF_NEG        0   0   0   0
CONNECT_POS_DUT       0   0   0   0
CONNECT_NEG_DUT       0   0   0   0
CAL_POS_DUT           0   0   0   0
CAL_NEG_DUT           0   0   0   0
CONNECT_PPMU_POS_DUT  0   0   0   0
CONNECT_PPMU_NEG_DUT  0   0   0   0
CAL_POS_TERM          0   0   0   0
CAL_NEG_TERM          0   0   0   0
CAL_POS_BYPASS        0   0   0   0
CAL_NEG_BYPASS        0   0   0   0
NOTCH_HF              0   0   0   0
ATTEN_HF_BIT3         0   0   0   0
ATTEN_HF_BIT3_GND     0   0   0   0
ATTEN_HF_BIT2         0   0   0   0
ATTEN_HF_BIT2_GND     0   0   0   0
ATTEN_HF_BIT1         0   0   0   0
```

ATTEN_HF_BIT1_GND	0	0	0	0
ATTEN_HF_BIT0	0	0	0	0
ATTEN_HF_BIT0_GND	0	0	0	0
LPF_HR_1KHZ	0	0	0	0
LPF_HR_1KHZ_BYPASS	0	0	0	0
ATTEN_HR1_0DB	0	0	0	0
ATTEN_HR1_3DB	0	0	0	0
ATTEN_HR1_6DB	0	0	0	0
ATTEN_HR2_0DB	0	0	0	0
ATTEN_HR2_9DB	0	0	0	0
ATTEN_HR2_18DB	0	0	0	0

**************** End of Dump Relay Counter for UltraSource ******************

You can also insert calls to count relay switches and to generate reports into a test program.

For more information, see Pins.RelayCounter in the VBT Syntax Reference.

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UltraPAC80 Source Programming : Using the BBAC to UltraPAC Converter Tool

Using the BBAC to UltraPAC Converter Tool

The BBAC to UltraPAC Converter Tool (BBAC2UltraPAC) allows you to convert BBAC test programs (for Source and Capture) to UltraPAC80 test programs (UltraSource and UltraCapture). The tool uses a graphical interface that allows you to:

- Select the test program to convert
- Select pattern files for conversion
- Convert .xls files to .ascii
- Select a pin map and a simulated configuration file
- Select the output file for the conversion
- View the progress of the conversion
- Review the conversion in a log file
- Create a new UltraPAC program that requires minimal work to complete

This section describes the use of the BBAC2UltraPAC tool. The following topics are available:

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UltraPAC80 Source Programming : Using the BBAC to UltraPAC Converter Tool : Starting the Conversion Tool

Starting the Conversion Tool

Use of this tool requires:

- A validated BBAC Source or BBAC Capture test program
- IG-XL V7.40 or later

Before starting the conversion, make sure no instances of IG-XL or Excel are running.

To start the BBAC to UltraPAC Conversion Tool:

1. Select Start>Run.

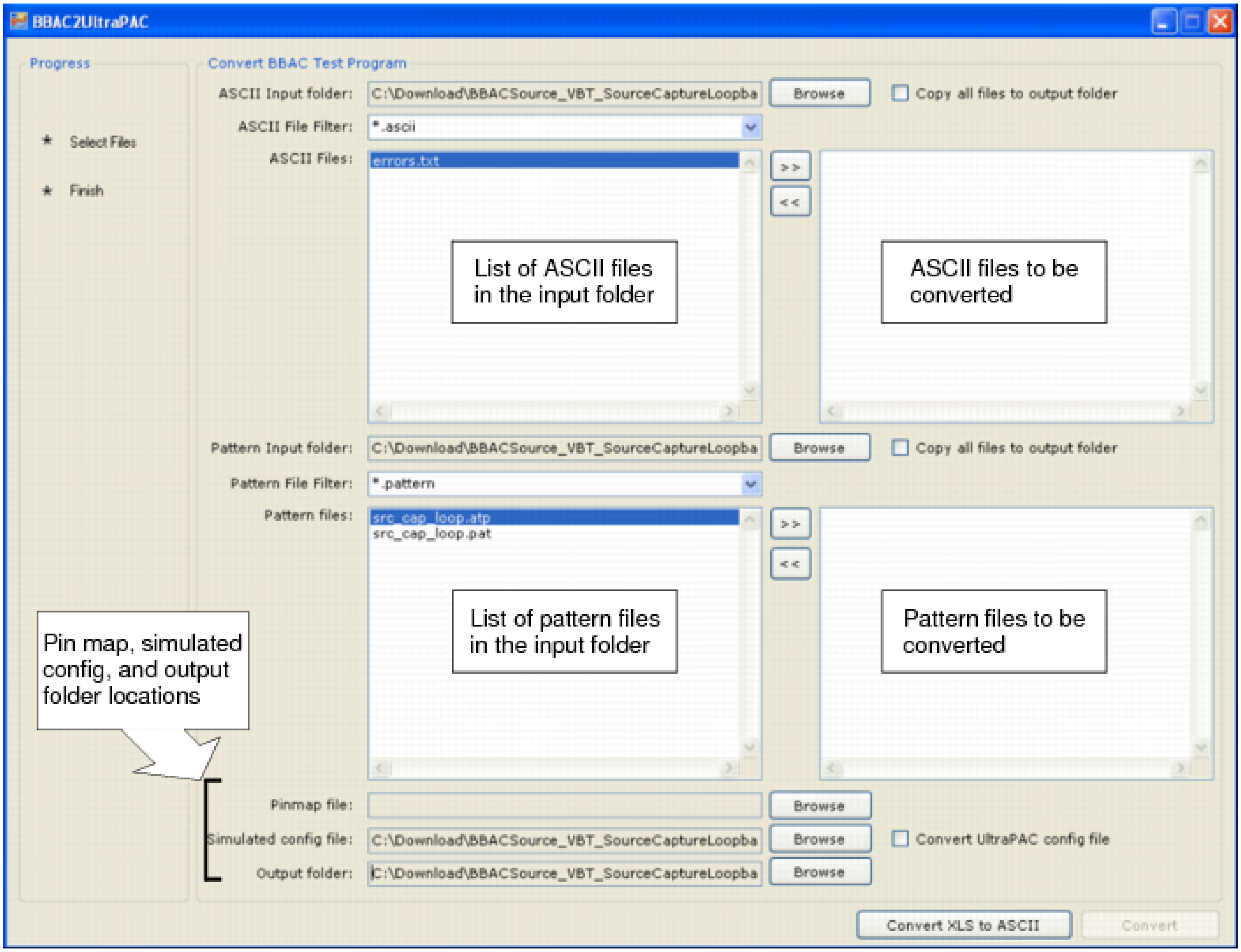
The Run window appears.

2. Enter bbac2ultrapac and click OK.

A Browser window appears.

3. Select the top folder for the test program you want to convert.

This opens the BBAC to UltraPAC Converter main window. This window is shown below with a sample program selected.



BBAC to UltraPAC Converter Main Window

4.	If the list of ASCII files in the input folder does not include ASCII files, click Convert XLS to ASCII. This brings up the Convert the XLS to ASCII TP window. This window allows you to convert the .xls files in your test program to .ascii files.

In this window, do the following:

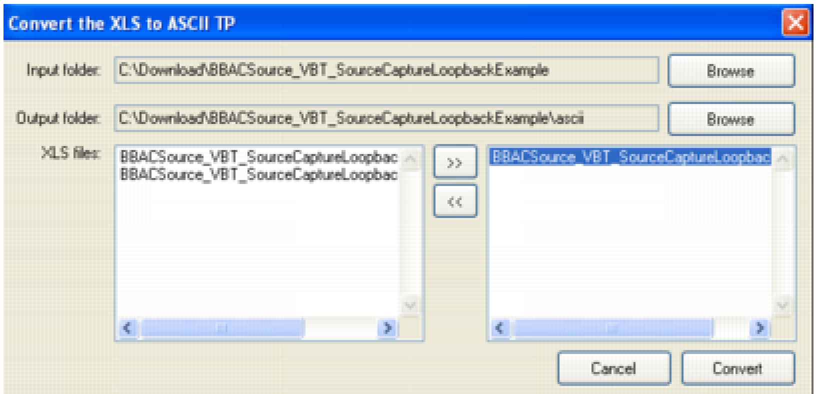
a. Click Browse and choose an input folder.

b. Click Browse to select an output folder, and create a folder called ascii.

c. Use the arrow button (>>) to select the .xls file to convert.

d. Click Convert.

When you click Convert, the tool opens the test program, saves the selected files to ascii, and closes automatically. This can take some time. During the conversion, you see a progress bar.



Convert XLS to ASCII TP Window

The next step is to select the test program files for conversion.

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UltraPAC80 Source Programming : Using the BBAC to UltraPAC Converter Tool : Selecting and Converting Test Program Files

Selecting and Converting Test Program Files

Once you have the ascii files you need, you need to select the files for conversion. Do the following the in converter main window:

1. Select the input folders for ASCII files and patterns:

   a. Click the Browse button next to ASCII Input folder and select the input folder.

   The list of files in the folder that can be converted appears in the ASCII Files field.

   b. Click the Browse button next to Pattern Input folder and select the input folder.

   The list of files in the folder that can be converted appears in the Pattern Files field.

2. Select the ASCII files to be converted:

   a. Select all the ascii test program files in the ASCII Files field and click >>.

   The files move to the right hand window.

   b. Check Copy all files to output folder.

   This puts ascii files that the converter does not process in the output folder. This makes loading the converted test program easier.

3.	Select the pattern files to be converted:

a.	Select the pattern files to be converted in the Pattern Files field and click >>.

The files move to the right hand window.
Only select pattern files that contain AC (BBAC) microcodes and avoid converting large digital only patterns (such as functional and scan).

b.	Check Copy all files to output folder.

This puts pattern files that the converter does not process in the output folder, retaining the original patterns folder structure.

4.	Click the Browse button next to Pinmap file and select a pin map file.

The pin map file must be an ASCII text worksheet.

5.	Click the Browse button next to Simulated config file. Select a Simulated Config file.

6.	Check Convert UltraPAC config file.

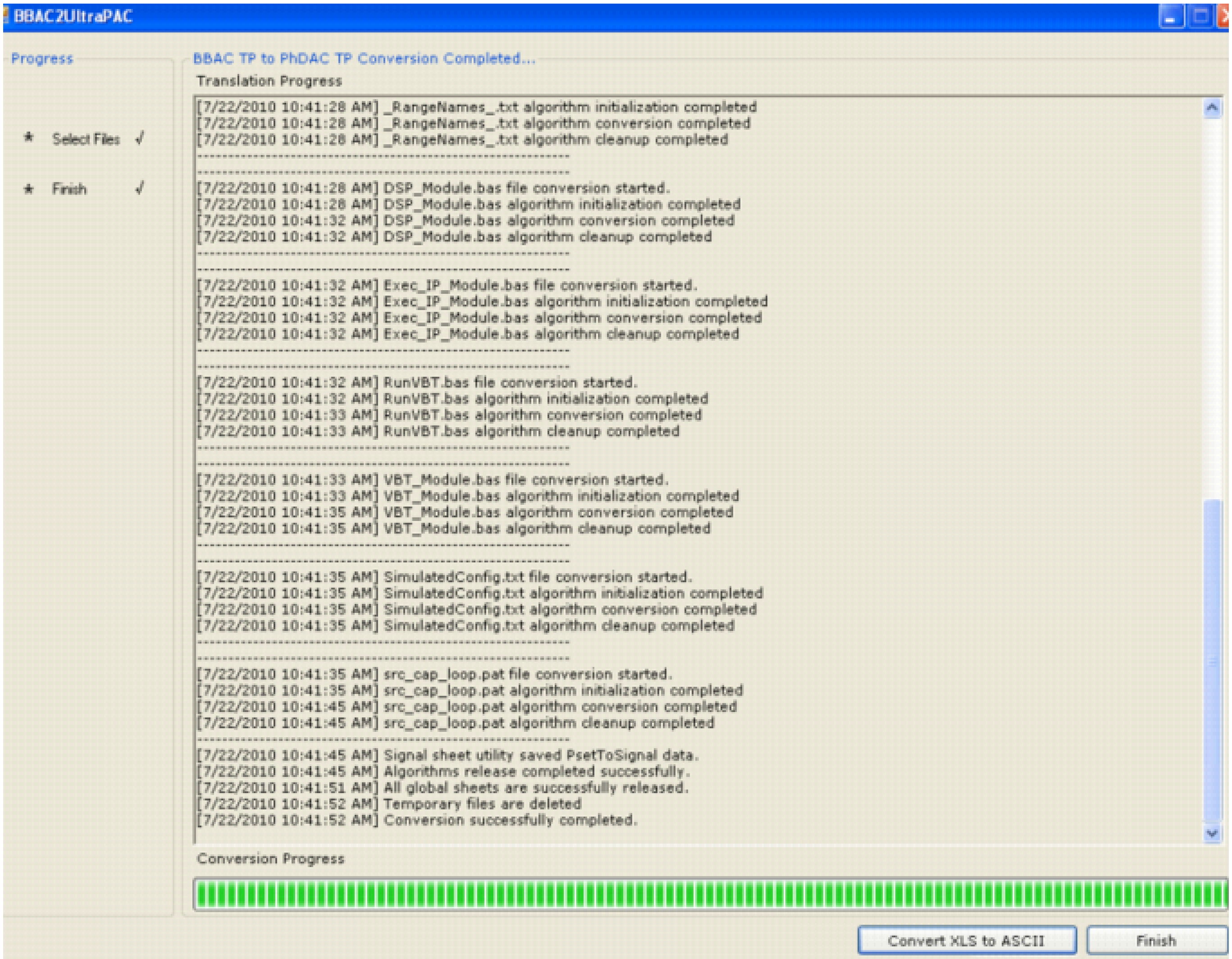
This control converts the existing BBAC config file to an UltraPAC config file. All instances of BBAC instrument in the simulated config will be converted to UltraPAC instruments, and a new simulated config file will be placed in the output folder.

7.	Click the Browse button next to Output folder and select the output folder for the converted test program files.

By default, the tool creates a folder called Output in the current folder.

8.	Click Convert.

The tool begins converting the file. You see a log of the conversion in the main window and a progress bar, as shown below. When the conversion is successfully completed, the Convert button changes to Finish.



Converter Main Window with Conversion Complete

9.	Click Finish to complete the conversion.

This closes the conversion tool. Sometimes for large test programs, the converter tool will appear to hang for several seconds on this step while temporary files are cleaned up. Wait and allow the tool to finish completely.

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UltraPAC80 Source Programming : Using the BBAC to UltraPAC Converter Tool : Opening the Converted Program and Completing Conversion

Opening the Converted Program and Completing Conversion

When the tool completes conversion, the Output folder contains the test program ASCII files converted to UltraPAC format. However, the program will usually require additional work to complete the conversion. This is because the tool cannot operate on some files or perform certain conversions.

Therefore, the final part of BBAC to UltraPAC conversion involves opening the completed test program and doing any manual adjustments. The procedure is as follows:

1. Copy any test program add-ins or utilities to the Output folder.

The Conversion utility does not convert or move any binary files (other than .pat files). If binary files are required, manually copy them to the output folder. An example would be IGXL_Utils.xla or DSP custom library .dll files.

2. Select Start>Programs>Teradyne IG-XL Vx.xx.xx_uflx>Open Existing Test Program.

This brings up the Open Existing IG-XL Test Program window.

3. In the conversion Output folder, select all the ascii files and click Open.

The converted UltraPAC test program should load and open.

4. Examine the converted VBT code in the test program.

Some BBAC VBT commands cannot be directly converted to UltraPAC or are obsolete. For these commands, the converter tool comments them and adds the comment “Manual assistance needed.” Examples are:

```
'Manual assistance needed.
'    TheHdw.BBACSource(SrcPin_I & ", " & SrcPin_Q).Connect tlBBACSourceFromREF,
tlBBACSourceToCCC
'Manual assistance needed.
'    .FlatnessCorrection = tlBBACCaptureFlatnessCorrectionNone
'Manual assistance needed.
'    .DitherEnabled = True
```

Perform a Find operation in VBT code using this string and manually inspect the code.

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UltraPAC80 Source Programming : Using the BBAC to UltraPAC Converter Tool : Details of the Conversion

Details of the Conversion

The following are details of the BBAC to UltraPAC conversion using the BBAC2UltraPAC tool:

- Sheet Conversions
- Pattern File Conversions
- Other Changes That May Require Manual Intervention

Sheet Conversions

The BBAC2UltraPAC tool converts several sheets and sheet columns.

For the Channel Map sheet:

Channel Map Conversion

BBAC			UltraPAC		
BBACSrcRef	##.d411		N/C (not supported in UltraPAC		
BBACSrcPos	##.b413	##.f413	UltraSource	##.SrcPos1	##.SrcPos2
BBACSrcNeg	##.b412	##.f412	UltraSource	##.SrcNeg1	##.SrcNeg2
BBACCapPos	##.b405	##.f405	UltraCapture	##.CapPos1A	##.CapPos2A
BBACCapNeg	##.b404	##.f404	UltraCapture	##.CapPos1A	##.CapPos2A

For other sheets, the conversion is:

BBAC2UltraPAC Sheet Conversions

Sheet	Conversion
Mixed Signal Timing (Type column)	• BBACCapture to UltraCapture • BBACSource to UltraSource
Signals	Follow VBT conversion
PSets	BBAC instrument PSet row converted to Signal sheet row.

Pattern File Conversions

The BBAC2UltraPAC tool converts pattern files as follows:

- Pattern headers are changed to UltraPAC by changing the instrument.
- Pattern body is changed as per the following table.

BBAC Source Microcode Conversions

BBAC Microcode	UltraPAC Microcode
PSet <PSetName>	LoadSettings <PSetName>
Resync	Resync
Start <SignalName>	Start <SignalName>
StartE <SignalName>	StartE <SignalName>
Start1 <SignalName>	Start1 <SignalName>
Start1E <SignalName>	Start1E <SignalName>
Stop	Stop
StopE	StopE

BBAC Capture Microcode Conversions

BBAC Microcode	UltraPAC Microcode
Trig <SignalName>	Trig <SignalName>
Resync	Resync
PSet <PSetName>	LoadSettings <PSetName>
Enable_Inst_Condition	Enable_Inst_Condition
Disable_Inst_Condition	Disable_Inst_Condition
Enable_Alarm	Enable_Alarm
Disable_Alarm	Disable_Alarm

Other Changes That May Require Manual Intervention

The following table lists BBAC features that the converter may not handle correctly. These items must be corrected manually.

Items Not Converted by BBAC2UltraPAC

Inconsistencies	Handled in the Translator
Amplitude range supported by BBAC but larger than that supported by UltraPAC.	Set the level to the max of the UltraPAC and return a warning/error in the output window and log file.
Sample rate supported by BBAC but larger than that supported by UltraPAC.	Set the level to the max of the UltraPAC and return a warning/error in the output window and log file.
Use of BBAC PDE procedures.	PDE procedures are not supported.
Uses incremental PSets.	Converting PSet to Signal requires incremental Signal. The PSet will have a name that you should be able to keep track of the settings.
Use of BBAC “Raw” language.	Conversion is not supported. Comment out the code and note in log and output window.
BBAC Source common mode voltage larger than allowed on UltraPAC Source.	Set the level to the max of the UltraPAC and return a warning/error in the output window and log file.
BBAC Capture dc baseline removal voltage larger than allowed on UltraPAC Capture.	Set the level to the max of the UltraPAC and return a warning/error in the output window and

	log file.
Use of DSP PDE procedures to analyze BBAC capture data.	PDE procedures are not supported.
Use of BBAC “AC Coupling.”	Conversion is not supported. Comment out the code and note in log and output window.
Use of BBAC “PreCharge.”	Conversion is not supported. Comment out the code and note in log and output window.
Use of BBAC SrcRef pins.	Ref Channel sheet.
Use of BBAC “External Trigger.”	Conversion is not supported. Comment out the code and note in log and output window.
Use of DIB access connections.	Conversion is not supported. Comment out the code and note in log and output window.
Use of BBAC “Dither.”	Conversion is not supported. Comment out the code and note in log and output window.
Use of BBAC “Flatness Correction.”	Conversion is not supported. Comment out the code and note in log and output window.
Disconnect and Connect statements to DGS.	Conversion is not supported. Comment out the code and note in log and output window.
Disconnect and Connect statements to CCC.	Conversion is not supported. Comment out the code and note in log and output window.
Use of specific voltage range not supported by the UltraPAC but within UltraPAC legal range.	Set the level to the max of the UltraPAC and return a warning/error in the output window and log file.
Use of PSets (UltraPAC uses only Signals language).	Convert to Signal based following VBT convention.

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UltraPAC80 Source Theory of Operation

UltraPAC80 Source Theory of Operation

This section provides conceptual and reference information on the UltraPAC80 Source instrument's hardware and signal path. The following topics are available:

- UltraPAC80 Instrument Board
- UltraPAC80 Source Signal Path
- Connectivity Options
- Advanced Features
- User View of DIB Pad Patterns

For information on UltraPAC80 Source instrument performance specifications and standards, see UltraPAC80 Specifications.

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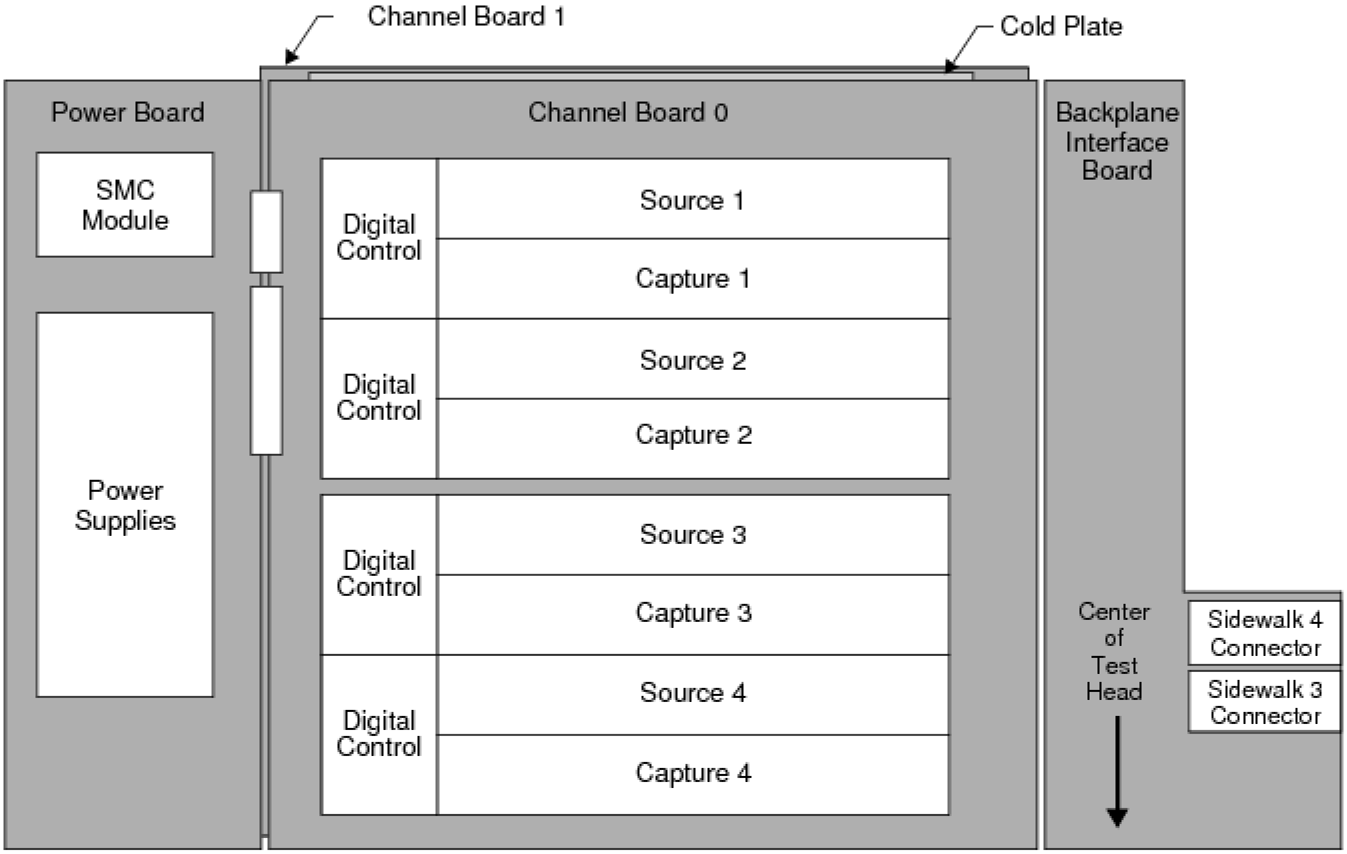
UltraPAC80 Source Theory of Operation : UltraPAC80 Instrument Board

UltraPAC80 Instrument Board

The UltraPAC80 instrument board assembly occupies a single slot in the UltraFLEX test head and consists of:

- Two identical channel boards mounted on opposite sides of a liquid-cooled cold plate.
  - Each channel board has four UltraSource channels and four UltraCapture channels, for a total of eight source and eight capture per instrument board
  - Each UltraSource and UltraCapture channel consists of a Pos pin, a Neg pin, a PPMU resource, DGS, and a converter (ADC for UltraCapture and DAC for UltraSource)
  - Each UltraSource and UltraCapture channel has a high frequency (HF) and a high resolution (HR) path
- A backplane interface board (BIB) that provides connections to the UltraFLEX backplane and manages signal flow between the test system and instrument boards
- A power board that converts the +48V system power to the various on-board power voltages that the UltraPAC80 board requires

The following figure shows the layout of an UltraPAC80 instrument board.



UltraPAC80 Instrument Board Hardware Block Diagram

UltraPAC Channel Boards

The UltraPAC channel boards contain the primary UltraPAC80 instrument control and tester communication components as well as the analog input and output stage of the UltraPAC80 instruments. Each channel board also has the following features:

- An interface to communicate with the databus of the tester computer
- 32 Pin Parametric Measurement Unit (PPMU) instruments, which can be used for parametric testing. Each UltraSource and UltraCapture instrument has two PPMU pins, one for each positive and each negative pin.
- MoveLink connections from the UltraPAC80 Capture channels to one or more dedicated DSP computers in the test system for high speed transfer and DSP processing of acquired capture data
- Instrument Sync-Link (ISL) connections from each UltraPAC80 channel to the Pattern Generator in the High Speed Digital instrument for accurate and repeatable synchronization

- A differential internal analog loopback connection shared between a pair of UltraPAC Source and a pair of UltraPAC Capture channels
- Four device ground sense (DGS) connections shared by the source, capture, and PPMU circuits for a pair of source and a pair of capture instruments

UltraSource and UltraCapture Channels

Each UltraPAC80 instrument board has eight UltraSource channels and eight UltraCapture channels in total. An UltraSource channel can drive AWG signals with bandwidths from DC to 80MHz to device analog input pins. An UltraCapture channel can capture signals from device output pins with bandwidths from DC to 75MHz.

Eight digital control FPGAs, four on each channel board, support the source and capture hardware. Each FPGA supports an UltraSource/UltraCapture channel pair.

Each UltraSource and UltraCapture channel also includes the following resources:

- Precision 24-bit high resolution AWG source and capture
- 16-bit high speed AWG source and capture
- Gain ranging
- Filtering
- Digital sample rate conversion
- Source and Capture and Signals memory
- Connection to PPMU per input/output pin

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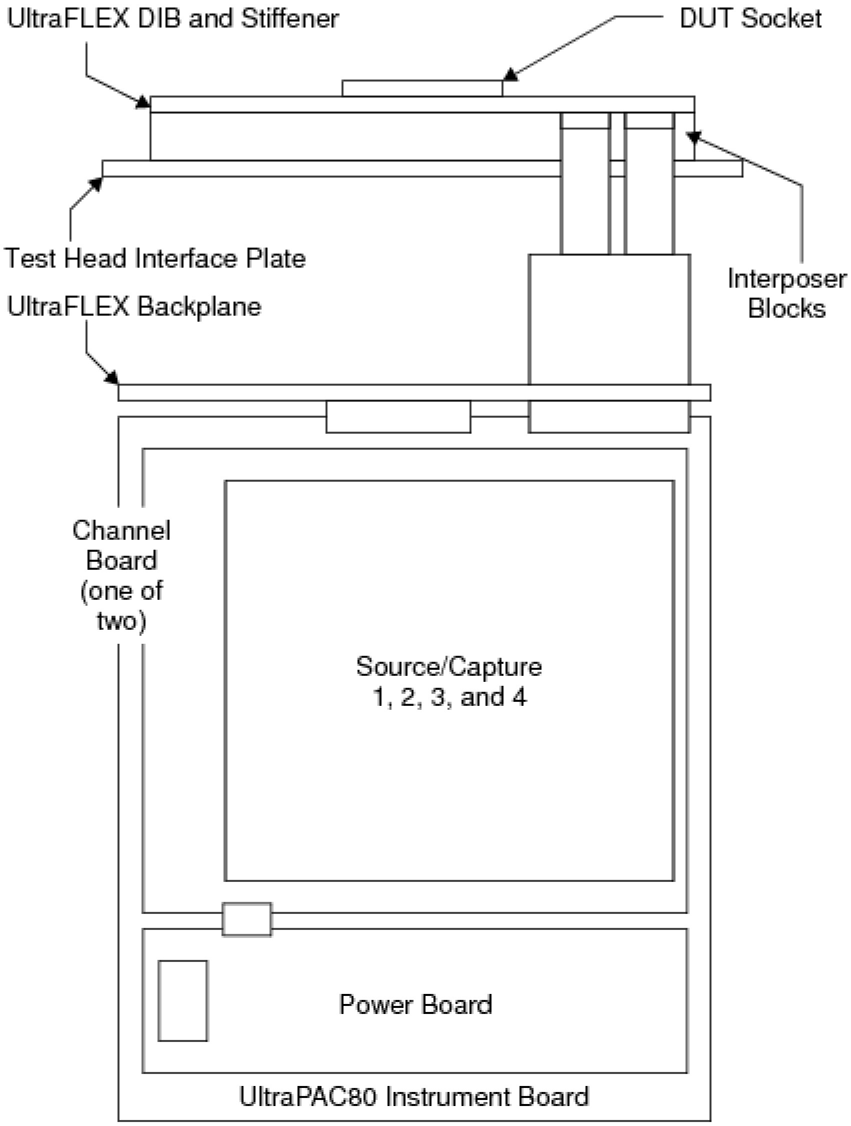
UltraPAC80 Source Theory of Operation : UltraPAC80 Instrument Board : Test Head Connections

Test Head Connections

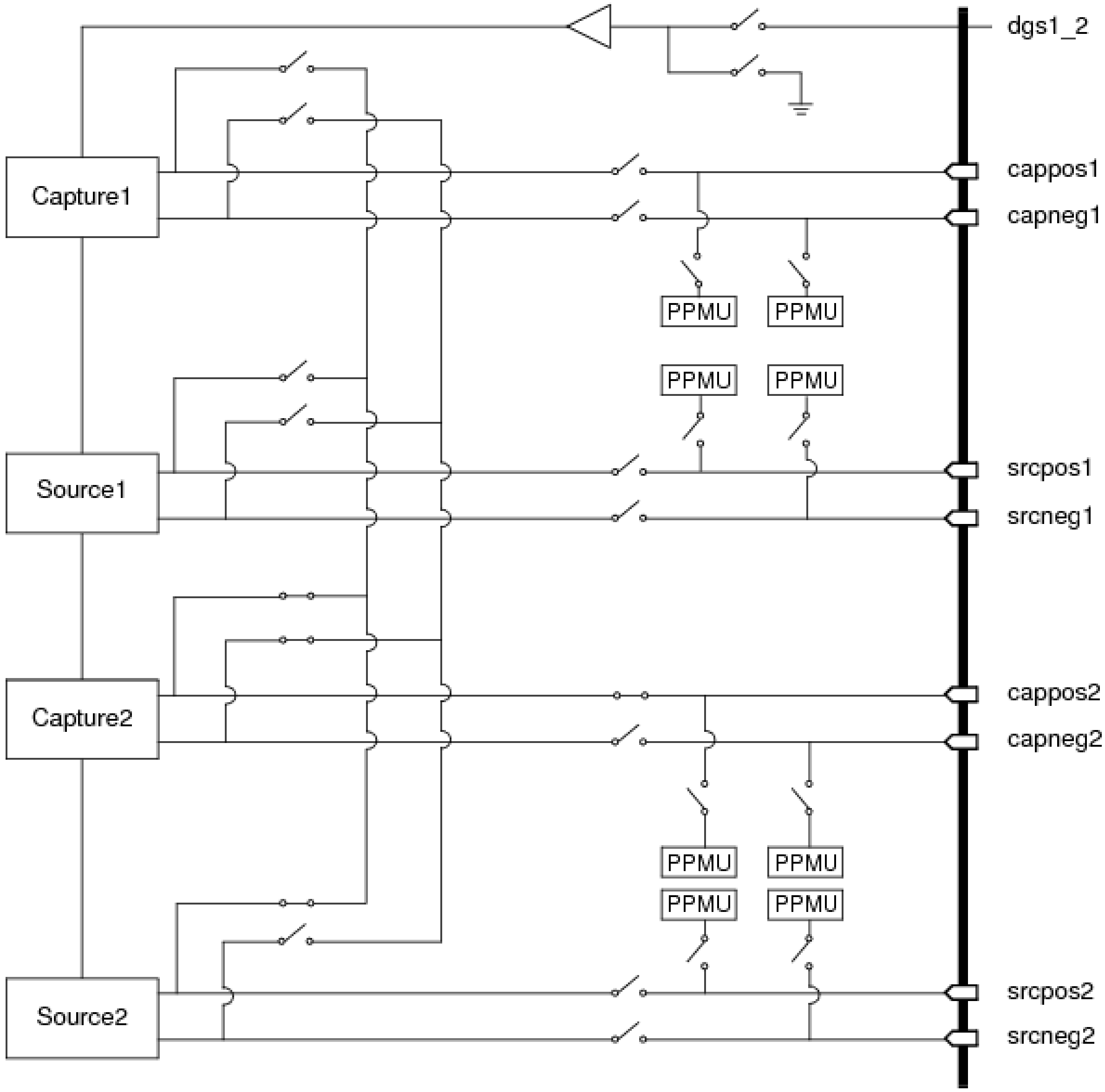
This section describes the connections of UltraPAC boards to the UltraFLEX test head backplane. The following figure shows how the instrument board is connected through interposer blocks to deliver power to the DUT.



UltraPAC80 Instrument Board Connection to the DUT

Up to eight UltraPAC80 boards can be installed in an UltraFLEX test head. An UltraPAC80 can be installed in any available slot in the test head. Channels are connected to the DIB using two interposer blocks occupying sidewalks 3 and 4 of the loadboard end.

The following figure shows a connection model for the UltraPAC80 signals to the DUT.



UltraPAC80 Channel Connections to the DUT

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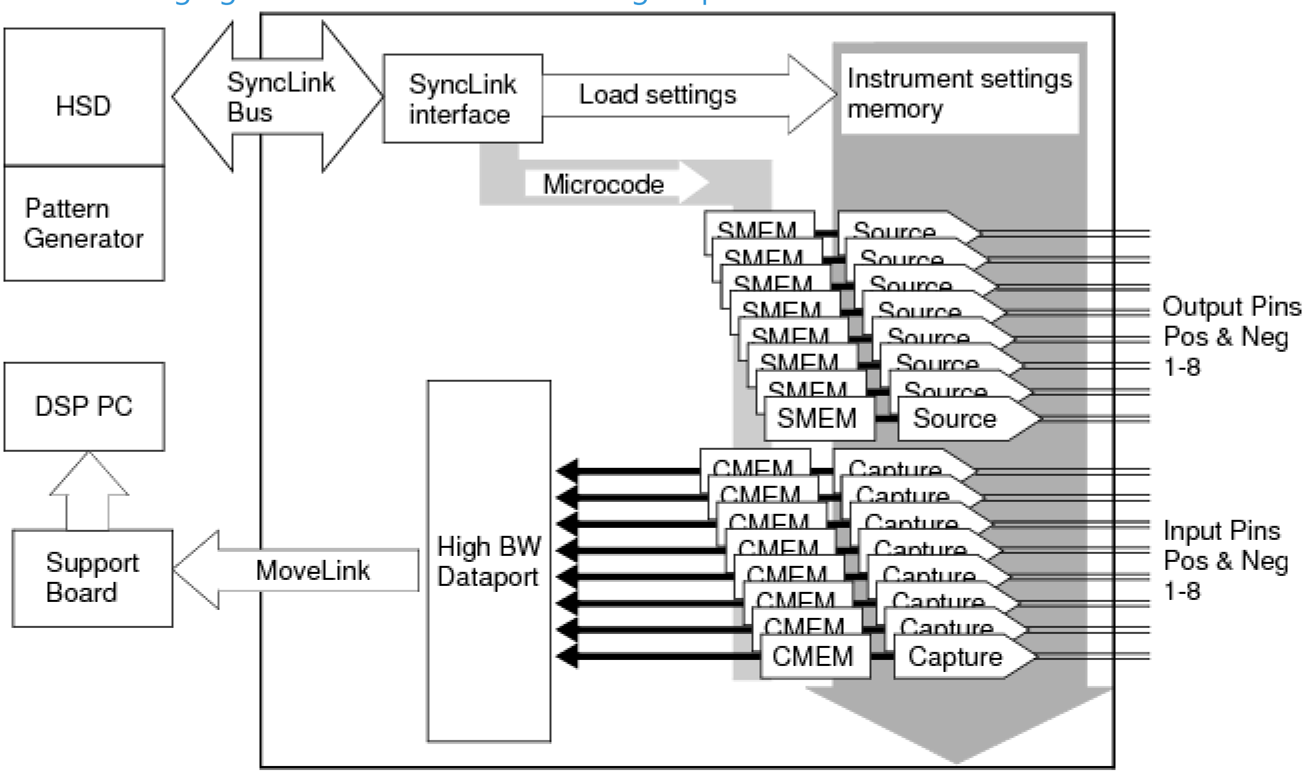
UltraPAC80 Source Theory of Operation : UltraPAC80 Source Signal Path

UltraPAC80 Source Signal Path

This section describes the architecture of an UltraSource channel. Each of the eight Source channels on the UltraPAC80 instrument board is identical.

The following figure shows the UltraPAC80 signal paths at the board level.



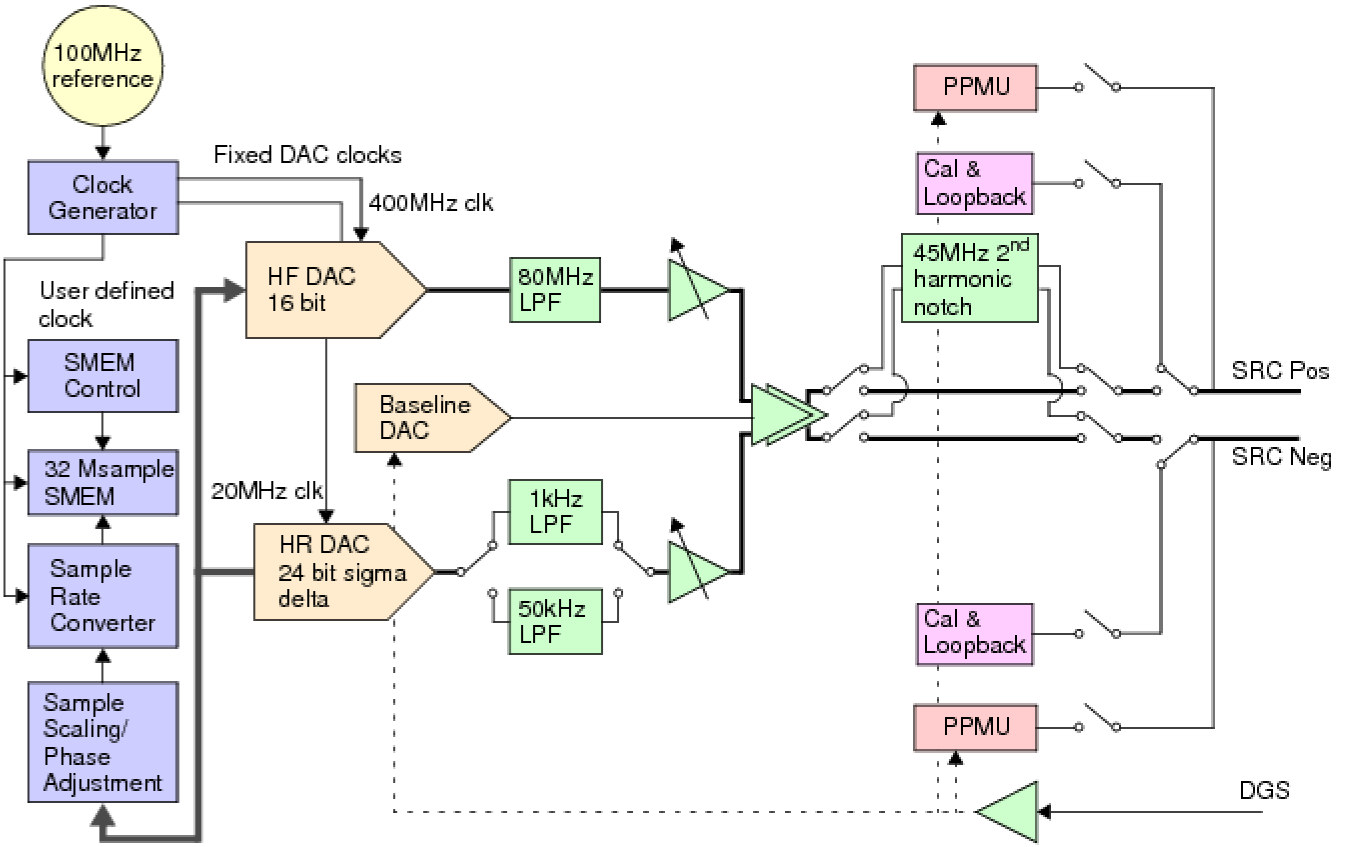
UltraPAC80 Board Diagram

The UltraPAC80 instrument includes eight independent UltraSources, each with separate high resolution (HR) and high frequency (HF) signal paths. The HR path covers the frequency range of DC - 50kHz, while the HF path operates from DC - 80MHz.

The use of two separate signal paths allows UltraSources to address a diverse set of tests over a wide range of speed and performance requirements. The high resolution (HR) path uses 24-bit based DACs and is used for low frequency source applications such as audio, static (ramp) linearity measurements, and precision lower speed analog-to-digital converters. The high frequency (HF) path uses 16-bit DACs and is used for general purpose sourcing for high speed ADC, video ADCs, wireless baseband, and baseband converters.

The figure below is a block diagram of a single UltraPAC80 Source channel. The following sections describe individual components of the block diagram.

Note that the 45MHz 2nd harmonic notch in the diagram is a filter used for very low distortion 45MHz single tone generation.



UltraPAC80 Source Channel Block Diagram (one of eight shown)

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UltraPAC80 Source Theory of Operation : UltraPAC80 Source Signal Path : Clock Generator

Clock Generator

The UltraPAC contains a clock generator circuit (one circuit common to all channels), which generates fixed clocks from the test system 100MHz master reference clock. These clocks are used to drive both the HF and HR source DACs. The clocks are not user programmable. User defined sample rates are achieved using sample rate conversion. (See Signal Processing.)

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UltraPAC80 Source Theory of Operation : UltraPAC80 Source Signal Path : Source Memory and Control (SMEM)

Source Memory and Control (SMEM)
User defined sample sets are stored in Source Memory (SMEM). The instrument then uses this data to source AWG waveforms from the source channel DACs. In default operation, when a source segment reaches its last sample, it will then jump to the first sample and continue sourcing. In this way, a continuous waveform can be sourced by automatically looping a segment.
The UltraPAC80 Source memory is 32M samples deep (33554432 samples per channel) and the largest single segment length is 32M, the full memory depth. IG-XL loads source segments as required by host computer and pattern starts. At any one time, up to 500 individual segments can be loaded in SMEM.

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UltraPAC80 Source Theory of Operation : UltraPAC80 Source Signal Path : Signal Processing

Signal Processing

The UltraPAC80 Source uses a series of digital signal processing stages to prepare the data for conversion by the DACs. First, the data is resampled to convert the sample rate from the user-specified sample rate to the instrument's fixed sample rate using a sample rate conversion algorithm. This process occurs in real time as sample are read from SMEM and provided to the DACs.

A benefit of sample rate conversion is that the sample rate that you define is mathematically determined and, therefore, provides a fine degree of control. Also, the DACs use a fixed sample frequency that is often much higher than the sample frequency you request. Therefore, the instrument operates in an oversampled mode and quantization noise is spread over a wider bandwidth and eventually removed by filters in the analog path.

The minimum programmable sample rate is 5ksps, and the maximum sample rate is 400Msps (HF mode) or 1Msps (HR mode). To ensure correct operation of the sample rate converter, however, the following rules determine the source sample rate and the number of samples in a source segment:

The minimum Fs allowed is:

- 10 x Fi (HR mode)

- 4 x Fi (HF mode)

where Fi is the highest frequency of interest.

The minimum N allowed is:

N = Maximum[20 or ceiling(1s x F_s)]

The 1s term in the above formula is necessary because the UltraSource memory controller requires at least 1s to elapse before the segment data can jump from its last sample back to its first sample.

The implication of a minimum segment length (time) of 1s it that sinusiodal waveforms with Fi greater than 1MHz must be generated with multiple cycles in the source segment. That is:
M >= Fi/1MHz

A digital amplitude scaling stage then adjusts the amplitude of the data depending on the requested amplitude and the currently selected hardware voltage range. You must set the peak amplitude of the signal to be sourced and choose the hardware voltage range (or use autorange). The digital amplitude scaling factor is automatically determined based on these factors. Calibration correction is also applied. An optionally programmable time-based skew is then applied to the waveform samples from -2500ps to 2500ps. You can use this skew can to apply custom generated calibrations, for example to remove skew due to DIB components.

For more information, see:

- Calculating Source Timing
- Mixed Signal Workshop

Clock Resolution

The UltraPAC80 has a period-based user programmable clock. Resolution is defined in terms of period resolution and therefore is frequency dependent. The UltraPAC80 frequency is governed by the equation:

$$\text{UltraPAC Clock Frequency} = \frac{100\ \text{MHz} \times 2^{44}}{M}$$

(2–1) $$\text{Where:} \quad M = [2^{42}, 2^{59} - 1]$$

The fine clock resolution allows the both the UltraSource and UltraCapture instruments to achieve any desired frequency to a degree of precision on a par with the digital subsystem. While the UltraPAC80 is not always guaranteed to be coherent with the digital subsystem in the strict sense (a perfect integer ratio relationship), any source or measurement error introduced due to coherence error is typically less than can be measured by the UltraPAC80 analog performance. You can use Mixed Signal Workshop to calculate strictly coherent clocking solutions while obeying sample rate and sample size rules for UltraPAC instruments.

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UltraPAC80 Source Theory of Operation : UltraPAC80 Source Signal Path : Source DACs

Source DACs

The UltraPAC source uses different DAC architectures for the HF and HR paths:

- The HF path uses 16-bit DACs. The key feature of this DAC is its high sample rate and bandwidth.
- The HR path uses 24-bit DACs. The key feature of this DAC is its good noise, SFDR, and THD performance, especially over the audio band. The HR DAC is also very linear for generating precise voltage ramps for static integral non linearity tests.

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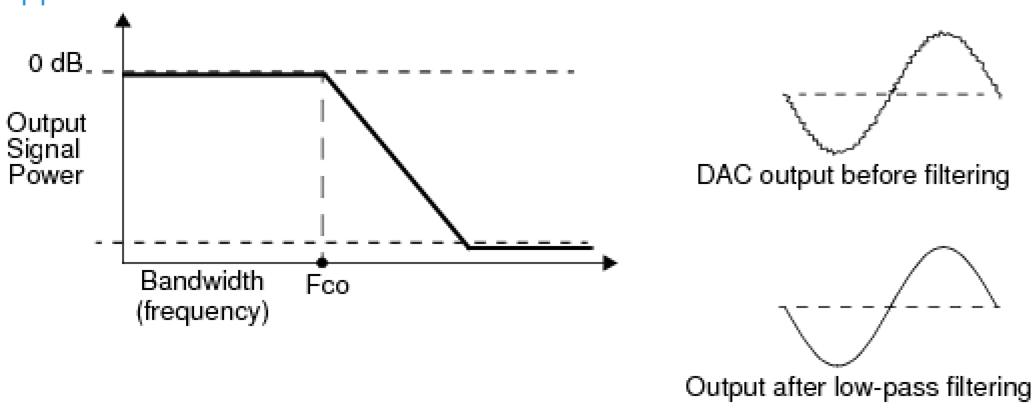
UltraPAC80 Source Theory of Operation : UltraPAC80 Source Signal Path : Filtering

Filtering

The source signal passes through a low-pass filter that smooths the output waveform by removing sampling images produced by the DACs. (The filter converts the stepped response of the DAC output samples to a smooth continuous waveform.) The filter also removes high frequency quantization noise from the synthesized signal due to the oversampling nature of the sample rate conversion architecture. The low-pass filter cutoff frequencies are 1kHz or 50kHz for the HR path (user selectable) and 80MHz for HF path. By specification, the filters are flat to the cutoff frequency. The cutoff frequency does not refer to the -3dB point.

Since the DACs always operate at a fixed clock rate, only one preset low-pass filter is required, with the pass band ending at one-half of the converter's clock rate. You can select a 1kHz LPF when using the HR path to generate low distortion 1kHz sinusoids for audio and precision ADC applications.



Low-Pass Filter Reduces High Frequency Quantization Noise

For filter specifications, see UltraPAC80 Specifications.

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UltraPAC80 Source Theory of Operation : UltraPAC80 Source Signal Path : Voltage Ranging

Voltage Ranging

The analog signal is amplified or attenuated by the range circuitry to achieve the programmed amplitude. A combination of digital scaling and hardware ranging is used to achieve the desired peak amplitude from an UltraSource without sacrificing dynamic range.

Voltage ranges are in 3dB increments. The HR mode has nine ranges, which are specified in units volts peak measured differentially. The range values are: 0.625, 0.884, 1.250, 1.768, 2.500, 3.536, and 5.000.

The HF mode has six ranges, which are specified in units volts peak measured differentially. The range values are: 0.530, 0.750, 1.061, 1.500, 2.121, and 3.000.

Voltage ranges are specified as units volts peak measured differentially across the positive and negative output pins. For single-ended device connections, only the positive or the negative pin is used. Therefore, the voltage output of an UltraSource is half the programmed voltage when used to source single-ended signals, since it only uses half of the differential pin pair.

For more information on UltraPAC80 Source compliance, see UltraPAC80 Specifications.

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UltraPAC80 Source Theory of Operation : UltraPAC80 Source Signal Path : Common Mode Voltage DAC

Common Mode Voltage DAC

The common mode voltage DAC is used to add a DC offset to the source DAC output voltage. This offset is applied to both the positive and negative outputs. This function is used to bias the UltraSource output to a device’s common mode bias voltage when sourcing differentially. You can also use this circuit to add DC offset when sourcing in single-ended configurations without adding DC offset to samples in source memory and using excessively high voltage ranges. Common mode voltage is applied relative to device ground sense (DGS.)

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UltraPAC80 Source Theory of Operation : UltraPAC80 Source Signal Path : Output Stage and Signal Delivery

Output Stage and Signal Delivery

The final output stage amplifiers sum the voltage from the source DAC and offset DAC, as well as DGS, and drive a differential output to the source positive and negative pins, ultrasrc_pos and ultrasrc_neg. The source output stage includes back-matched resistors in series between the output amplifier and signal delivery pins, providing the source with 50 ohms (per side) output resistance in HF mode and 1.5 ohm (per side) output resistance in HR mode.

Take care when sourcing to DUT inputs that are not high impedance, as this will cause a voltage drop across the resistor divider formed by the source output resistance and the device input resistance. All UltraSource programmed source voltages are open circuit and do not account for loading by the DIB or the DUT.

Signal delivery also includes DGS input pins, which are used to provide the device's ground voltage reference to the instrument card. All voltages sourced from UltraPAC are always the programmed level relative to the DGS voltage. DGS lines should typically be grounded close to the DUT, but they can be grounded near the interposer for low current applications. A single DGS input is shared amongst two source and two capture channels: 1-2, 3-4, 5-6, 7-8. DGS pin to channel routing is fixed and is not user selectable.

Each UltraSource positive, negative, and DGS line incorporates a corresponding shield. The shield is used to isolate the source signals from EMF interference as they pass through coaxial cabling from the instrument boards to the interposer interface at the DIB. The shields also carry return currents from the source lines.

Therefore, make sure to ground the shields on the DIB. The best practice for the grounding these shields is application dependent. Typically, they are grounded to the analog DIB ground plane near the interposer connections.

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UltraPAC80 Source Theory of Operation : Connectivity Options

Connectivity Options

The following topics describe the connectivity options for connecting a UltraPAC80 source instrument to resources through the DIB:

- Differential Device Input Connections
- Single-Ended Device Input Connections
- Mixing Differential and Single-Ended Output Connections
- Bidirectional Connections

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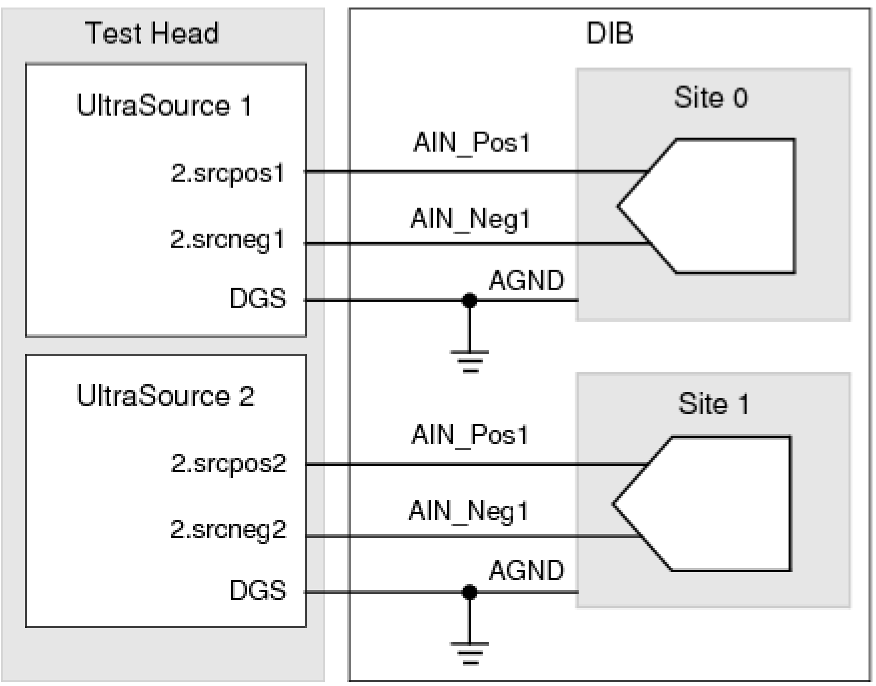
UltraPAC80 Source Theory of Operation : Connectivity Options : Differential Device Input Connections

Differential Device Input Connections

The UltraPAC80 Source reference pin can be connected to the DIB ground to make a differential connection between the UltraPAC80 Source instrument and a device with differential inputs.



UltraSource Differential Input Connections

Pin Map for UltraSource Differential Connections

Group Name	Pin Name	Type
	AIN_Pos1	Analog
	AIN_Neg1	Analog

Channel Map for UltraSource Differential Connections

Pin Name	Package Pin	Type	Site 0	Site 1
AIN_Pos1		UltraSource	2.srcpos1	2.srcpos2
AIN_Neg1		UltraSource	2.srcneg1	2.srcneg2

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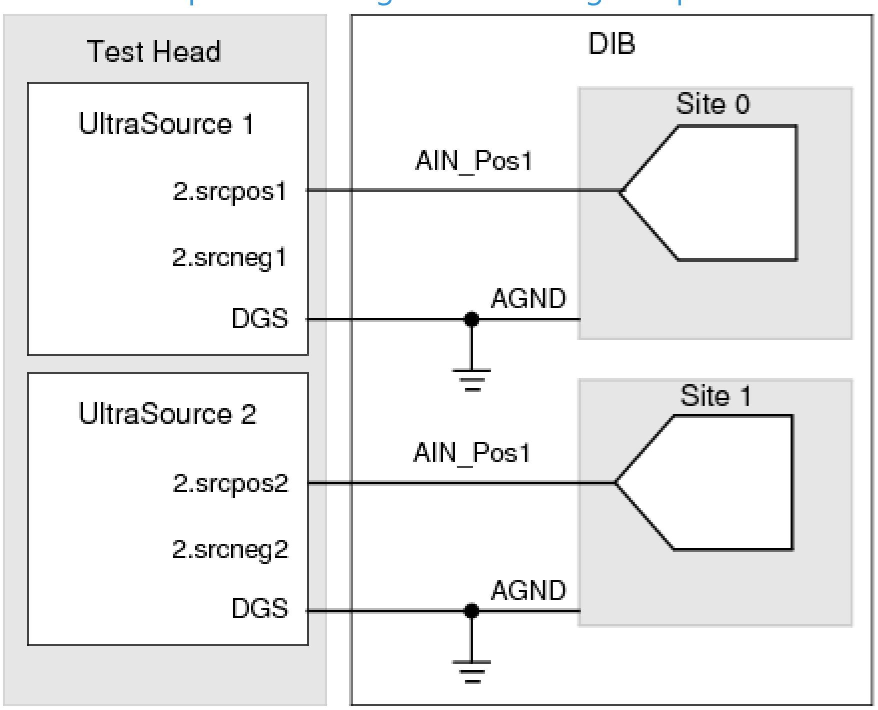
UltraPAC80 Source Theory of Operation : Connectivity Options : Single-Ended Device Input Connections

Single-Ended Device Input Connections

For connection to single-ended device inputs, DIB circuitry is not required for single-ended to differential conversion, as shown in the figure. To connect the UltraPAC80 Source instrument to single-ended devices, connect either the positive or negative pin of the UltraPAC80 differential pin pair to the single-ended device input. You need not ground the unused pin during the device test. It should remain disconnected. Do not connect the unused pin to ground, as this will short the channel’s output buffer to ground and degrade performance.



UltraSource Connections for Single-Ended DUT Pin Only

Pin Map for UltraSource Single-Ended Connection

Group Name	Pin Name	Type
	A_IN	Analog

Channel Map for UltraSource Single-Ended Connection

Pin Name	Package Pin	Type	Site 0	Site 1
AIN		UltraSource	2.srcpos1	2.srcpos2

You can use both the positive and negative pins of a common source instrument to source a waveform to individual single-ended device input pins on the same site or different sites. The specifications for the positive and negative pins are identical.

Note the following:

- If you connect the negative pin to drive a device input, then the source voltage is inverted relative to the positive pin and waveform samples.
- You cannot control the positive and negative pin’s AC signal amplitude and DC offset individually.

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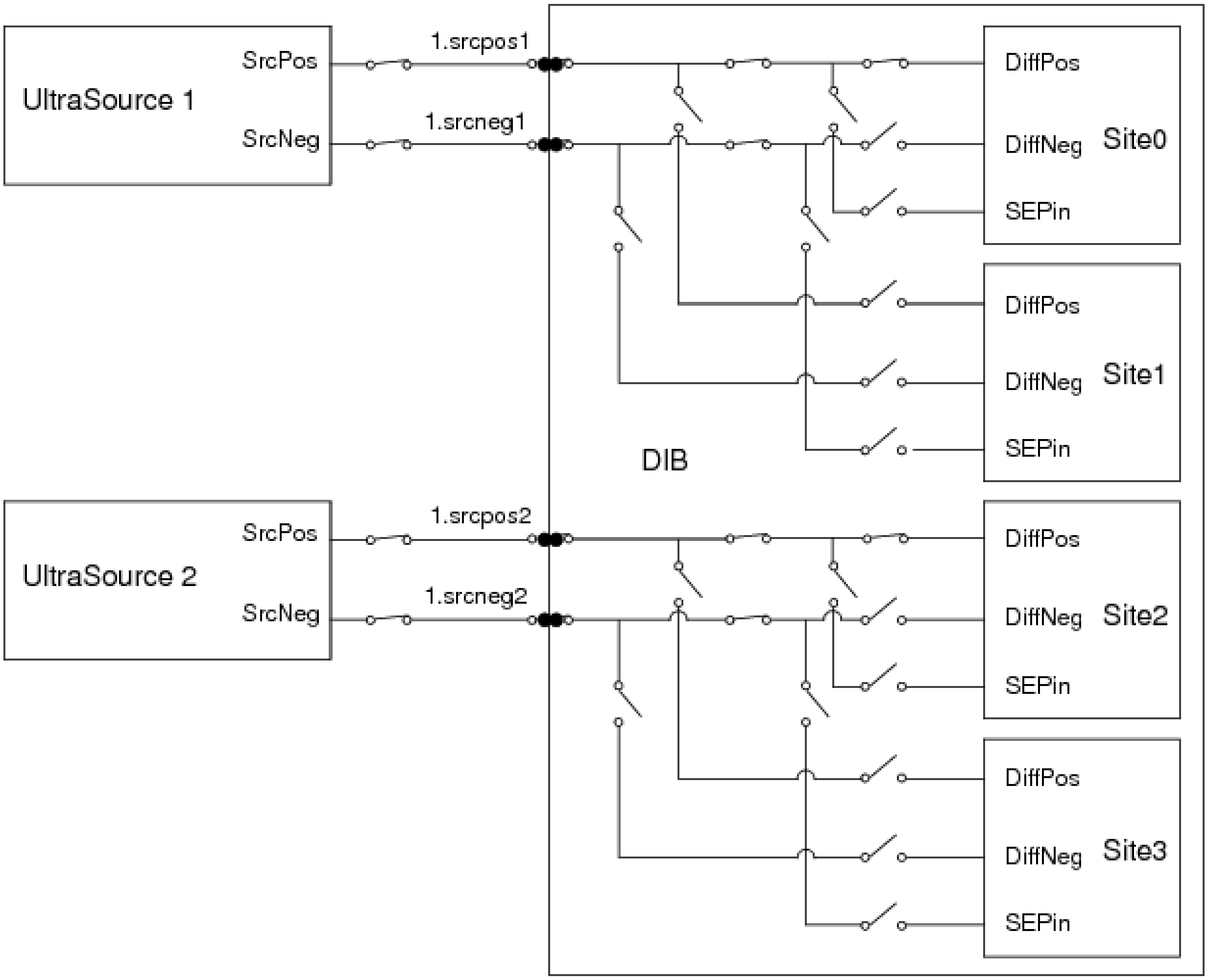
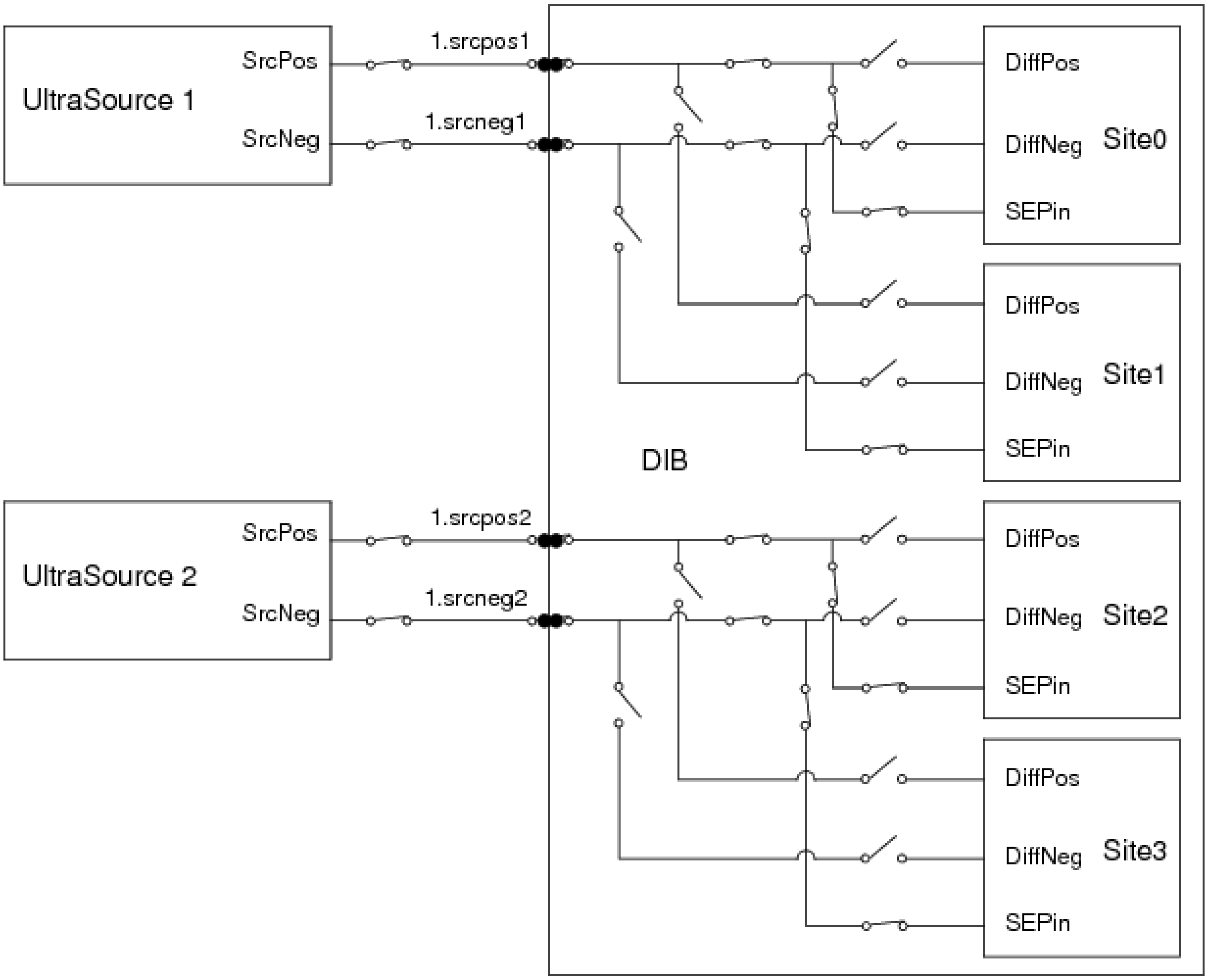
UltraPAC80 Source Theory of Operation : Connectivity Options : Mixing Differential and Single-Ended Output Connections

Mixing Differential and Single-Ended Output Connections

The following figures show a quadsite program using two UltraSource instruments in slot 1. Differential pins DiffPos and DiffNeg are shared across sites. UltraSource 1 is specified behind sites 0 and 1, and UltraSource 2 is specified behind sites 2 and 3. The same channels are shared by a single-ended pin, SEPin, using each channel (both Pos and Neg) of the UltraSource instruments. To share the two UltraSource among four sites, you need to enter a site loop on even sites and then odd sites.



Quadsite Differential Pins in Even Sites Connected to Two UltraSource Instruments



Quadsite Single-Ended Pins Connected to Two UltraSource Instruments

The following is the channel map associated with the example quad-site program.

Channel Map for Quadsite Differential and Single-Ended Pins Sharing Two UltraSources

Pin Name	Package Pin	Type	Site 0	Site 1	Site 2	Site 3
DiffPos		UltraSource	1.srcpos1	site0	1.srcpos2	site2
DiffNeg		UltraSource	1.srcneg1	site0	1.srcneg2	site2
SEPin		UltraSource	1.srcpos1	10.srcneg1	1.srcpos2	10.srcneg2

The VBT syntax to connect the differential pins would be:

TheHdw.UltraSource.Pins(“DiffPos,DiffNeg”).Connect

The VBT syntax to connect the single-ended pins would be:

TheHdw.UltraSource.Pins(“SEPin”).Connect

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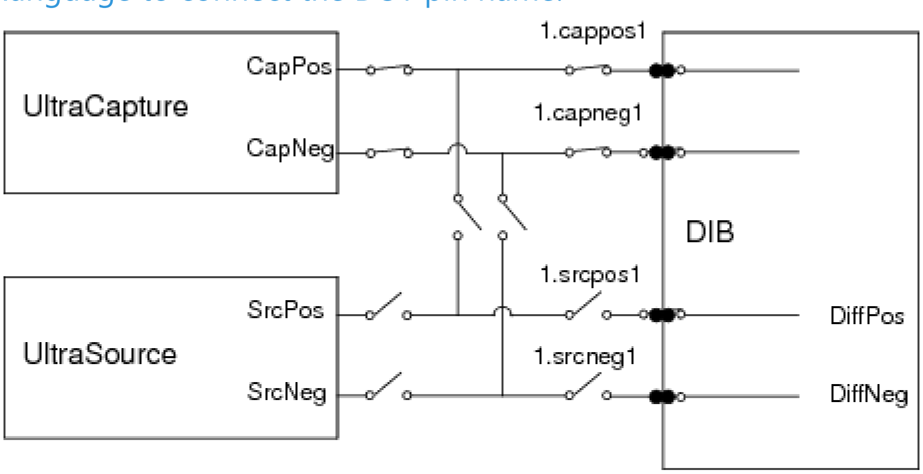
UltraPAC80 Source Theory of Operation : Connectivity Options : Bidirectional Connections

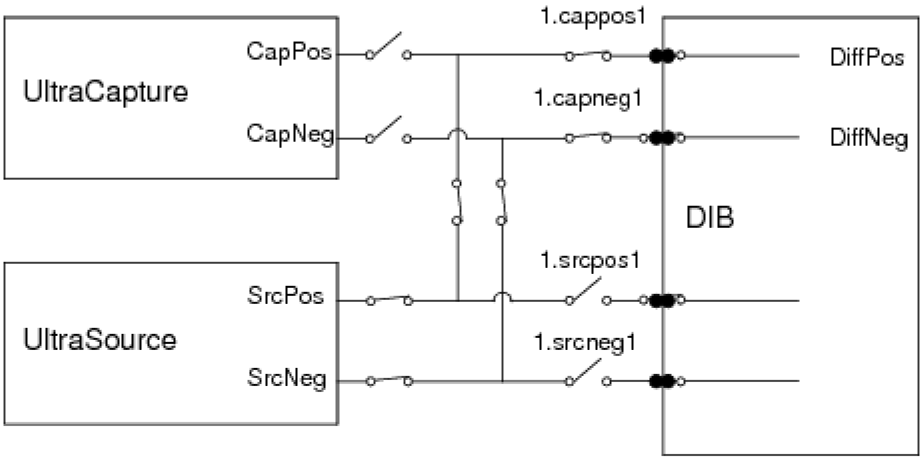
Bidirectional Connections

UltraPAC80 has bidirectional capability, which permits connections from an UltraSource or through the capture DUT connections only without using additional relays on the DIB. This feature allows you to source to a pair of DUT pins in one test and then capture from the same pair of DUT pins for the next test.

The following figures show the UltraCapture instrument associated with the same DUT pins, DiffPos and DiffNeg, as the UltraSource instrument. The VBT Connect statement connects the correct instrument to the DUT depending on whether you connect using UltraSource or UltraCapture language to connect the DUT pin name.



Bidirectional DiffPos and DiffNeg Pins, UltraCapture Connected



Bidirectional DiffPos and DiffNeg Pins, UltraSource Connected

The following is the channel map required to create bidirectional connections.

Channel Map for Multiple Resource Connections

Pin Name	Package Pin	Type	Site 0
DiffPos		UltraCapture	1.cappos1
DiffNeg		UltraCapture	1.capneg1
DiffPos:M		UltraSource	1.cappos1
DiffNeg:M		UltraSource	1.capneg1

The VBT syntax to connect the UltraSource would be:

TheHdw.UltraSource.Pins("DiffPos,DiffNeg").Connect

Note that only two of four instrument pairs (UltraSource and UltraCapture) can be used at the same time for bidirectional connections.

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UltraPAC80 Source Theory of Operation : Advanced Features

Advanced Features

The following topics are included:

- Signals Memory
- Connections Sets
- IQ Testing

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UltraPAC80 Source Theory of Operation : Advanced Features : Signals Memory

Signals Memory

UltraPAC80 instruments use a hardware-based state engine to store instrument settings and connections in memory. This is similar to the PSet memory for other UltraFLEX instruments.

The UltraPAC80 does not use the PSet programming model. Instead, signals that are defined in software are loaded and stored to hardware on execution of the .LoadSettings signal method. First, the state of the instrument settings is resolved and instrument settings for that state are stored to specialized signals memory in the UltraPAC hardware. Next, IG-XL sends an instruction to execute that setup. At any later time when a .LoadSettings is executed for this signal, if the signal has not changed, a command is issued to have the hardware execute the state change from signals memory instead of setting each hardware parameter one at a time. This feature provides fast instrument setup changes from VBT and from pattern microcode (LoadSettings pattern microcode).

Additionally, you can use the VBT language .ReloadSettings to apply previously defined signals setups without first resolving the software signal setup. The benefit of the .ReloadSettings command is that it executes even faster than .LoadSettings. However, you cannot use this command to load newly defined or altered signals definitions from software to hardware.

For more information, see:

- LoadSettings
- ReloadSettings
- UltraSource Pattern Microcodes

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UltraPAC80 Source Theory of Operation : Advanced Features : Connections Sets

Connections Sets

Connection sets (CSets) are connection state setups that can be defined using VBT programming and stored in specialized CSet memory in the UltraPAC hardware. Once defined and loaded to hardware memory, a state can be applied with a single command either from VBT or under pattern microcode control, thereby allowing for fast instrument connection changes.

A benefit of using CSets is that the UltraPAC need not be disconnected at the end of every test to ensure test code modularity. Using the legacy incremental programming model such as .Connect VBT language (which is supported), you must always know the current connection state to apply the appropriate sequence of connects and disconnects to get the instrument to the new desired state. This typically leads test programmers to always put the instrument in a known reset state (disconnected) at the end of every test to be able to effectively reorder test flow.

CSets present a state programming model, not an incremental programming model, hence you only need to set the correct CSet state at the beginning of a test and not worry about reset states to ensure modularity. When a CSet is applied, the UltraPAC hardware automatically resolves the connect state and throws appropriate relays on the board to get to that state. If the board is already in the appropriate state when a CSet is applied, then the hardware does nothing, thereby extending relay lifetime.

For more information, see Connection Sets.

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UltraPAC80 Source Theory of Operation : Advanced Features : IQ Testing

IQ Testing

The UltraPAC80 instrument can serve as a baseband I/Q signal generator for testing wireless devices. The I channel and Q channel must be programmed individually to make up the I/Q signal generating pair.

UltraPAC features include a focused calibration routine to align corresponding source pairs in phase and amplitude.

For more information, see Achieving IQ Alignment for UltraSource.

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UltraPAC80 Source Theory of Operation : User View of DIB Pad Patterns

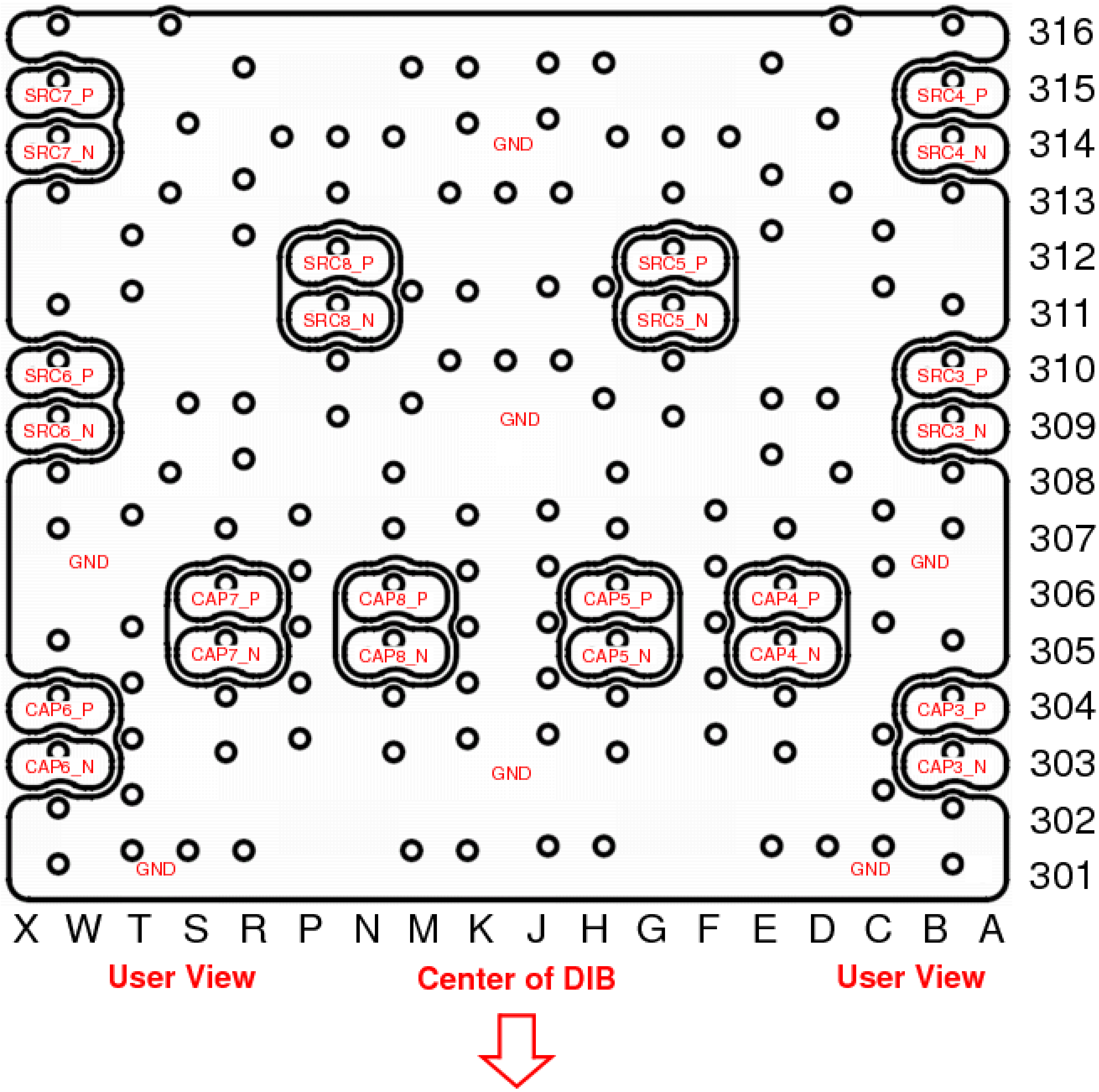
User View of DIB Pad Patterns

The following figures show the arrangement of UltraPAC80 board signals as seen from the user perspective, looking at the topside (device) of the DIB/HIB/PIB and towards the test system. The UltraPAC pinout will work with BBAC and TurboAC pad patterns. However, not all TurboAC and BBAC features are supported on UltraPAC80.

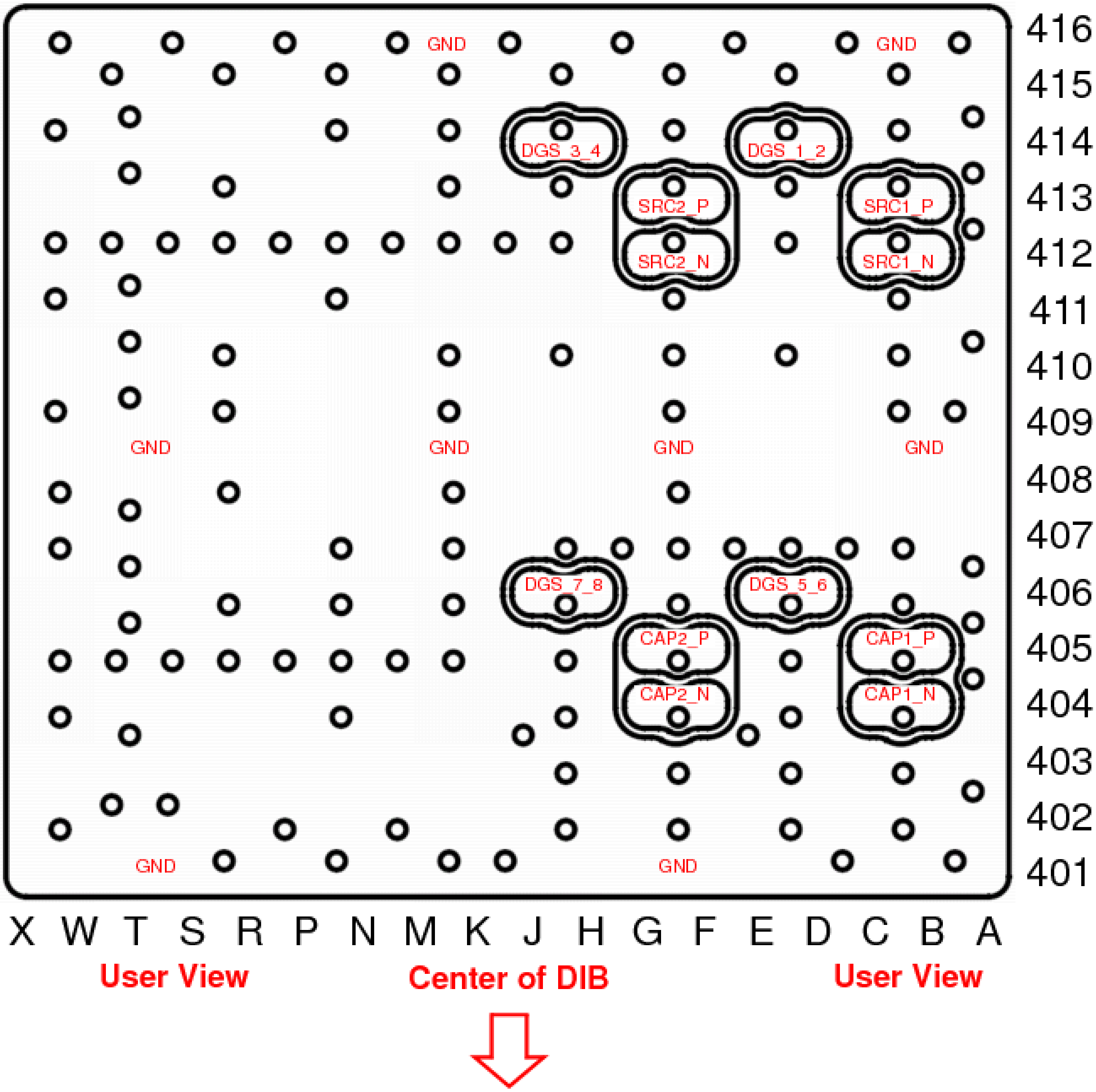
☑ **Note:**

When designing a DIB, use the tester view of the pad patterns (looking at the stiffener side of the DIB/HIB/PIB) because all the artwork and design kits use that view. The UltraPAC80 pad pattern is shown in the UltraPAC80 Signal Pad Pattern (Tester View) section of the UltraFLEX DIB Design Guidelines.

The following figures show the UltraPAC80 user view of the pad patterns for sidewalks 3 and 4 respectively.



UltraPAC80 Pad Pattern for Sidewalk 3 (User View)



UltraPAC80 Pad Pattern for Sidewalk 4 (User View)

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